Optimization-Based Clinical Decision Support for Curable Hypertension

SVM: Hard-Margin Feasibility, Soft Margins, and Kernel Validation

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Code and data: `https://github.com/nttssv/convex_optimization_svm_example`

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1 Problem Identification

Primary aldosteronism (PA) is an important cause of secondary hypertension. The clinically relevant subtype decision is whether PA is unilateral or bilateral. Unilateral PA can often be treated surgically, whereas bilateral PA is usually managed medically [3]. The aim of this project is to formulate this binary classification problem as a maximum-margin optimization problem and to determine whether a linear hard-margin support vector machine (SVM) can be trained on the available project data.

The analysis begins with a *linear hard-margin* SVM to determine whether a perfect linear separator exists. That feasibility audit uses no slack variables, finite penalty parameter, hinge-loss penalty, or kernel transformation. Once the audit proves that the classes are not linearly separable, the report introduces a linear soft-margin SVM, derives its primal and dual formulations, and selects its penalty parameter C by leakage-safe grid search. The resulting sequence is therefore hard-margin feasibility, soft-margin optimization, nested cross-validated performance, and an RBF kernel comparison with explicit overfitting diagnostics.

2 Dataset and Feature Representation

The source report describes a cohort of 114 patients evaluated for PA at Changi General Hospital [4]. Patient-level clinical data were not available for this course project, so the numerical analysis uses a synthetic CGH-like cohort with 114 observations: 56 labelled unilateral PA (UPA) and 58 labelled bilateral PA (BPA). The synthetic data are used to test the optimization workflow and are not evidence of clinical performance.

Ten continuous predictors are used: plasma aldosterone concentration, plasma renin activity, potassium, tumor size, 18-hydroxycortisol, 18-oxocortisol, systolic blood pressure, diastolic blood pressure, antihypertensive burden, and age. Let $x_i \in \mathbb{R}^p$ denote the feature vector for observation i, with $p = 10$, and let

$$t_i = \begin{cases} +1, & \text{for UPA,} \\ -1, & \text{for BPA.} \end{cases} \tag{2.1}$$

The four positively skewed hormone measurements are log-transformed. Within each cross-validation split, medians, means, and standard deviations are estimated from the training fold only; these training-fold quantities are then used for imputation and standardization of the held-out fold.

3 Descriptive Statistics of the Predictors

Before any margin model is fitted, each predictor is summarized on its raw measurement scale. Table 1 reports completeness, central tendency, dispersion, range, and skewness for the ten continuous features across all 114 observations. No predictor contains a missing value in this synthetic cohort.

The skew (log) column is the empirical justification for the log transformation. The six non-log transformed predictors are close to symmetric, with skewness between -0.19 and 0.22, whereas the four remain assays are strongly right-skewed: 1.45 for PAC, 1.75 for PRA, 1.95 for 18-OHF, and 3.81 for 18-oxoF. After the log transformation all four fall within $[-0.20, 0.25]$. Only the four log-transformed hormones are transformed, and the decision follows from their marginal distributions alone, not from any outcome-dependent criterion.

Table 1: Descriptive statistics for the ten continuous predictors on the raw measurement scale. IQR is the interquartile range and Range is the observed minimum to maximum. Skew is the adjusted Fisher–Pearson skewness; Skew (log) repeats it after the $\log(x + 10^{-6})$ transformation and is shown only for the four log-transformed hormone assays.

Predictor	Mean	Median	IQR	Range	Skew	Skew (log)
PAC	26.53	22.55	14.65–35.15	4.70–99.60	+1.45	-0.20
PRA	0.74	0.55	0.35–0.89	0.07–3.03	+1.75	-0.14
Potassium	3.59	3.60	3.23–3.90	2.60–4.70	+0.18	–
Tumor size	13.39	13.15	9.60–16.38	0.00–28.50	+0.22	–
18-OHF	1404.11	1105.50	692.75–1705.00	180.00–5994.00	+1.95	+0.14
18-oxoF	109.26	63.80	44.08–134.38	7.70–952.40	+3.81	+0.25
Systolic BP	139.42	141.00	130.00–149.75	100.00–193.00	+0.16	–
Diastolic BP	88.13	89.00	78.25–95.75	60.00–120.00	+0.13	–
DDD	2.31	2.00	2.00–3.00	0.00–5.00	-0.04	–
Age	49.36	49.00	43.00–56.75	21.00–72.00	-0.19	–

Table 2 splits each predictor by subtype and reports two complementary quantities: the standardized mean difference (Cohen's d) and the Welch t-statistic. They answer different questions. Cohen's d measures *how large* the gap between the subtype means is, expressed in pooled standard deviations, and does not change with cohort size. The Welch statistic measures *how precisely* that gap is resolved: it divides the same difference by its standard error, so it grows in proportion to $\sqrt{m}$.

Table 2: Predictor distributions by PA subtype. Cohen's d is the standardized mean difference (UPA minus BPA) using the pooled standard deviation; Welch t is the same difference divided by its standard error.

Predictor	UPA mean (SD)	BPA mean (SD)	Cohen's d	Welch t	p
PAC	34.9 (17.5)	18.4 (9.5)	+1.17	+6.20	$<$0.001
PRA	0.5 (0.4)	1.0 (0.7)	-0.80	-4.32	$<$0.001
Potassium	3.4 (0.4)	3.8 (0.5)	-0.95	-5.09	$<$0.001
Tumor size	15.4 (4.8)	11.4 (5.9)	+0.74	+3.98	$<$0.001
18-OHF	1883.0 (1225.7)	941.8 (557.0)	+0.99	+5.25	$<$0.001
18-oxoF	148.1 (151.2)	71.8 (66.9)	+0.66	+3.46	$<$0.001
Systolic BP	141.4 (16.3)	137.5 (17.5)	+0.23	+1.22	0.224
Diastolic BP	92.3 (10.5)	84.1 (11.3)	+0.75	+4.03	$<$0.001
DDD	2.5 (0.9)	2.1 (1.0)	+0.36	+1.94	0.055
Age	48.6 (11.0)	50.1 (9.6)	-0.15	-0.80	0.425

These contrasts are descriptive. Computed on the full cohort, they are not used to select features: every model in this report is fitted on all ten predictors inside the cross-validation pipeline. More importantly, a large univariate effect does not imply linear separability - d compares class means, whereas the hard-margin constraint of Section 4 requires *every individual observation* to lie on the correct side of one hyperplane. Figure 1 shows that even PAC, the strongest single predictor, leaves the two subtypes heavily overlapped.

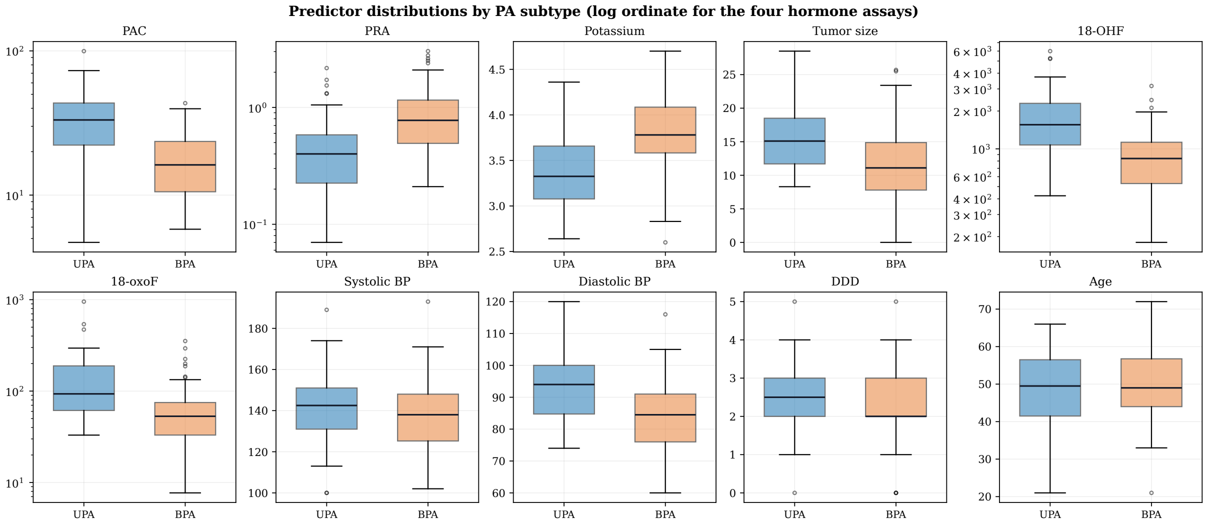

Figure 1: Distribution of each predictor by PA subtype. Boxes span the interquartile range with the median marked; whiskers extend to 1.5×IQR and points beyond are drawn individually. The four log transform features (PAC, PRA, 18-OHF, and 18-oxoF) use a logarithmic ordinate to make their right-skewed distributions legible. Every panel shows overlapping subtype distributions, which is why no single feature - and, as Section 4 shows, no linear combination of all ten - can separate the classes perfectly.

4 Hard-Margin SVM: Primal, Dual, and Existence

4.1 Primal Optimization Problem

The linear decision score is $s_i = w^\top x_i + b$, where $w \in \mathbb{R}^p$ is the normal vector of the separating hyperplane and $b \in \mathbb{R}$ is the intercept. The hard-margin primal SVM is the convex quadratic program [1, 6, 5]

$$\begin{aligned} \min_{w,b} \quad & \phi(w,b) = \frac{1}{2}\|w\|_2^2 \\ \text{subject to} \quad & t_i(w^\top x_i + b) \geq 1, \qquad i = 1, \dots, m. \end{aligned} \tag{4.1}$$

The constraints require every training observation to be correctly classified with functional margin at least one. Minimizing $\|w\|_2^2/2$ maximizes the geometric margin $2/\|w\|_2$ between the two supporting hyperplanes [6].

In a hard-margin SVM there is no per-observation loss term to differentiate. Margin violations are forbidden by constraints rather than assigned a finite loss. Consequently, the quantity differentiated below is the *primal objective*, together with the affine constraint functions.

4.2 First and Second Derivatives

Define the augmented parameter vector, quadratic matrix, and signed augmented feature vector as

$$\theta = \begin{bmatrix} w \\ b \end{bmatrix}, \qquad Q = \begin{bmatrix} I_p & 0 \\ 0 & 0 \end{bmatrix}, \qquad u_i = t_i \begin{bmatrix} x_i \\ 1 \end{bmatrix}. \tag{4.2}$$

The objective and constraints can then be written as

$$\phi(\theta) = \frac{1}{2}\theta^\top Q \theta, \qquad g_i(\theta) = 1 - u_i^\top \theta \leq 0. \tag{4.3}$$

The first derivative of the objective is

$$\nabla\phi(\theta) = Q\theta = \begin{bmatrix} w \\ 0 \end{bmatrix}, \tag{4.4}$$

and its second derivative is the constant Hessian

$$\nabla^2\phi(\theta) = Q = \begin{bmatrix} I_p & 0 \\ 0 & 0 \end{bmatrix}. \tag{4.5}$$

For each margin constraint,

$$\nabla g_i(\theta) = -u_i = -t_i \begin{bmatrix} x_i \\ 1 \end{bmatrix}, \qquad \nabla^2 g_i(\theta) = 0. \tag{4.6}$$

Because $Q \succeq 0$ and every g_i is affine, the primal problem is convex [5]. The objective is strictly convex in w but not in the complete vector (w, b) because the intercept is unpenalized. Thus the optimal w is unique whenever a solution exists, whereas the intercept need not be unique. There is also no hinge boundary or nondifferentiable point in this constrained primal formulation.

Section 6.2 evaluates $u_i^\top\theta$ row by row in a ten-observation spreadsheet illustration.

4.3 Lagrangian and KKT Conditions

The derivatives of the primal objective describe only how $\|w\|_2^2/2$ changes; they do not by themselves enforce the margin constraints. Indeed, setting the objective gradient to zero would give $w = 0$, which is generally infeasible. The Lagrangian is therefore introduced as an auxiliary function-not as a replacement for the primal objective-to combine the objective with the constraints and express constrained optimality through the KKT conditions.

Introduce one multiplier $\alpha_i \geq 0$ for each inequality constraint. The Lagrangian is

$$\mathcal{L}(w, b, \alpha) = \frac{1}{2}\|w\|_2^2 + \sum_{i=1}^{m} \alpha_i \left[1 - t_i(w^\top x_i + b)\right]. \tag{4.7}$$

Its first derivatives with respect to the primal variables are

$$\nabla_w \mathcal{L} = w - \sum_{i=1}^{m} \alpha_i t_i x_i, \qquad \frac{\partial \mathcal{L}}{\partial b} = -\sum_{i=1}^{m} \alpha_i t_i. \tag{4.8}$$

At a stationary point these derivatives vanish, giving $w = \sum_i \alpha_i t_i x_i$ and $\sum_i \alpha_i t_i = 0$. Thus the separating direction is determined by the training observations whose constraints are active and whose multipliers are positive, rather than by the unconstrained minimum $w = 0$.

For fixed α, the Hessian with respect to (w, b) is again

$$\nabla^2_{(w,b)} \mathcal{L} = \begin{bmatrix} I_p & 0 \\ 0 & 0 \end{bmatrix}. \tag{4.9}$$

When the primal is feasible, its Karush–Kuhn–Tucker (KKT) conditions are necessary and sufficient for global optimality [2, 8]:

$$\begin{aligned} w - \sum_{i=1}^{m} \alpha_i t_i x_i = 0, &\quad \sum_{i=1}^{m} \alpha_i t_i = 0, \\ t_i(w^\top x_i + b) - 1 \geq 0, &\quad \alpha_i \geq 0, \\ \alpha_i \left[t_i(w^\top x_i + b) - 1\right] = 0, &\quad i = 1, \ldots, m. \end{aligned} \tag{4.10}$$

The final equality is complementary slackness. Read in both directions it states that an inactive constraint forces a zero multiplier ($t_i(w^\top x_i + b) - 1 > 0 \Rightarrow \alpha_i = 0$), while a positive multiplier forces an active constraint ($\alpha_i > 0 \Rightarrow t_i(w^\top x_i + b) = 1$); observations with $\alpha_i > 0$ therefore lie on a supporting margin and are the support vectors. Complementary slackness is also exactly the term that makes $\sum_i \alpha_i [t_i(w^\top x_i + b) - 1]$ vanish in the Lagrangian at optimality, which is what collapses the duality gap to $p^\star = d^\star$ [7]. Stationarity alone is not sufficient: primal feasibility, dual feasibility, and complementary slackness must also hold.

The KKT conditions are sufficient here because the objective is convex and the margin constraints are affine. They are also necessary whenever the hard-margin problem is feasible, for the reason established in Section 4.5. Conversely, if the data are not linearly separable, primal feasibility fails and no triplet (w, b, α) can satisfy the complete KKT system.

4.4 Dual Problem and the Meaning of the Dual Bound

The primal and dual optimize different variables and therefore point in opposite directions. The primal minimizes over the classifier parameters (w, b). For any fixed multiplier vector $\alpha \geq 0$, define the dual function as the smallest Lagrangian value attainable by the primal variables,

$$q(\alpha) = \inf_{w,b} \mathcal{L}(w, b, \alpha). \tag{4.11}$$

For every primal-feasible (w, b), the constraint terms $1 - t_i(w^\top x_i + b)$ are nonpositive. Hence

$$q(\alpha) \leq \mathcal{L}(w, b, \alpha) \leq \frac{1}{2}\|w\|_2^2. \tag{4.12}$$

Thus each dual-feasible α supplies a *lower bound* on every feasible primal objective value. This is the dual bound, the weak-duality inequality of [7]. The dual maximizes $q(\alpha)$ because it seeks the largest, and therefore tightest, lower bound; this explains why the primal is a minimization problem while the dual is a maximization problem.

Using stationarity, $w = \sum_i \alpha_i t_i x_i$ and $\sum_i \alpha_i t_i = 0$, eliminates w and b from the Lagrangian and gives

$$\begin{aligned} \max_{\alpha} \quad & q(\alpha) = \sum_{i=1}^{m} \alpha_i - \frac{1}{2} \sum_{i=1}^{m} \sum_{j=1}^{m} \alpha_i \alpha_j t_i t_j x_i^\top x_j \\ \text{subject to} \quad & \alpha_i \geq 0, \qquad \sum_{i=1}^{m} \alpha_i t_i = 0. \end{aligned} \tag{4.13}$$

Weak duality always gives $d^* \leq p^*$. Whether that inequality tightens to an equality is a separate question, settled by the constraint qualification of the next subsection.

4.5 Slater's Condition and Strong Duality

Weak duality is automatic, but the dual is a faithful stand-in for the primal only when the gap closes [7]. For a convex program

$$\begin{aligned} \min_{x} \quad & f(x) \\ \text{subject to} \quad & f_i(x) \leq 0, \qquad i = 1, \ldots, m, \\ & Ax = b, \end{aligned} \tag{4.14}$$

Slater's constraint qualification asks for a strictly feasible point: some $\tilde{x} \in \operatorname{int} \mathcal{D}$ with

$$f_i(\tilde{x}) < 0 \quad \text{for all } i = 1, \ldots, m, \qquad A\tilde{x} = b, \tag{4.15}$$

where A has full row rank. When it holds, $p^* = d^*$; and if $d^* > -\infty$, the dual optimum is attained [2].

Two refinements decide the matter here. The strict inequality is needed only for the *non-affine* inequality constraints, so for a convex program whose inequality constraints are all affine, Slater's condition collapses to ordinary feasibility.

That is precisely the present case. The objective $\frac{1}{2}\|w\|_2^2$ is convex and every margin constraint $1 - t_i(w^\top x_i + b) \leq 0$ is affine in (w, b), so strong duality holds for the hard-margin SVM as soon as one separating hyperplane exists. Feasibility is therefore the single hinge of the entire hard-margin analysis. The certificate of Section 6.1 shows the constraints are infeasible on this dataset, so $p^* = +\infty$ and there is no finite primal value for the dual to certify - which is why a dual solution must never be read as a trained classifier before feasibility is checked.

The soft-margin program of Section 5 differs in exactly the way that matters. Its constraints $1 - \xi_i - t_i(w^\top x_i + b) \leq 0$ and $-\xi_i \leq 0$ are again affine, and the point $(w, b, \xi) = (0, 0, \mathbf{1})$ is feasible for *any* dataset. Slater's condition therefore holds unconditionally, strong duality is guaranteed in advance, and the vanishing primal–dual gaps reported later (4.9×10^{-10} for the linear model, 2.2×10^{-11} for the RBF model) are genuine checks on solver accuracy rather than assumptions being taken on trust.

4.6 Linear Separability and Existence of a Hard-Margin Solution

Let $U \in \mathbb{R}^{m\times(p+1)}$ have rows $u_i^\top = t_i[x_i^\top, 1]$ and let $\theta = [w^\top, b]^\top$. A hard-margin solution exists if and only if the linear system

$$U\theta \geq \mathbf{1} \tag{4.16}$$

is feasible. This condition is also equivalent to strict linear separability: if some hyperplane correctly separates all observations with strictly positive signed scores, rescaling (w, b) makes the smallest signed score equal to one; conversely, any point satisfying the unit-margin inequalities is a strict separator.

The existence question is checked by the zero-objective linear program

$$\min_{\theta}\ 0 \qquad \text{subject to} \qquad -U\theta \leq -\mathbf{1}. \tag{4.17}$$

If it is feasible, its output supplies a valid starting point for minimizing the quadratic primal objective. If it is infeasible, changing the objective from zero to $\|w\|_2^2/2$ cannot make the empty feasible set nonempty: the hard-margin primal has no solution. This logical order separates two questions that are often confused: feasibility asks whether any separating hyperplane exists, whereas optimization selects the maximum-margin hyperplane only after existence has been established.

4.7 Worked Separable Primal–Lagrangian–Dual Example

This deliberately small example makes the primal, Lagrangian, and dual views visible before returning to the clinical feature space. It uses the five observations in Table 3. Each point is embedded in the same ten-dimensional notation as the main analysis by setting $x_3 = \cdots = x_{10} = 0$. The symbol y_i used in this example has exactly the same role as t_i elsewhere in the report. Numerical values in the table, solution, margin evaluation, multiplier calculation, dual check, and figure are displayed to two decimal places; exact integers are retained where they make the symbolic constraint algebra easier to read.

Table 3: Five-point teaching dataset embedded in ten feature dimensions.

Point	x_1	x_2	$x_3, \ldots, x_{10}$	y
A	-2.00	0.00	0.00	-1
B	-1.00	1.00	0.00	-1
C	1.00	1.00	0.00	$+1$
D	2.00	0.00	0.00	$+1$
E	2.00	2.00	0.00	$+1$

Geometric intuition. The closest opposite-class observations in the horizontal direction are B at $x_1 = -1.00$ and C at $x_1 = 1.00$. The widest symmetric separating strip therefore places its decision boundary halfway between them at $x_1 = 0.00$, with supporting margins at $x_1 = -1.00$ and $x_1 = 1.00$. Figure 2 shows this geometry. B and C touch the margins, while A, D, and E are farther from the decision boundary.

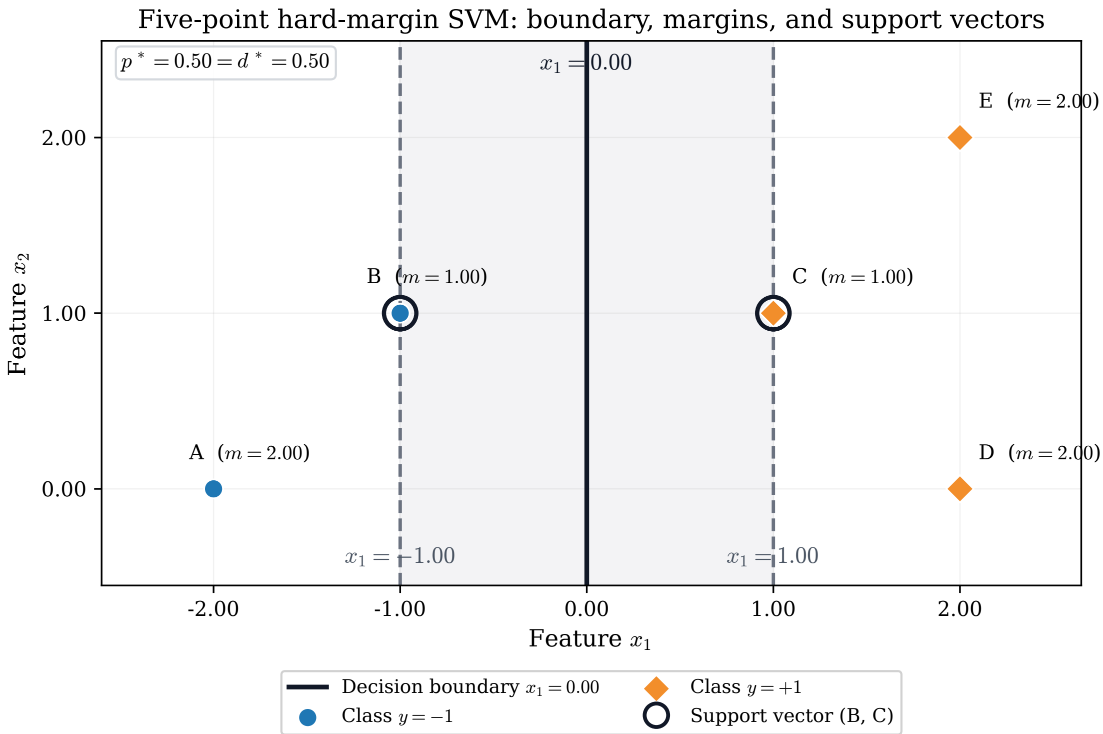

Figure 2: Five-point hard-margin example. The black line is the decision boundary, the dashed lines are the two supporting margins, and the rings identify support vectors B and C. Here $m_i = y_i(w^\top x_i + b)$ denotes the signed functional margin.

Primal problem and the five constraints. For $w = (w_1, \ldots, w_{10})^\top$, the hard-margin primal is

$$\begin{aligned} \min_{w,b} \quad & \frac{1}{2}\|w\|_2^2 = \frac{1}{2}\sum_{j=1}^{10} w_j^2 \\ \text{subject to} \quad & y_i(w^\top x_i + b) \geq 1, \qquad i \in \{A, B, C, D, E\}. \end{aligned} \tag{4.18}$$

Substituting every observation gives

$$\begin{aligned}
&\text{A:} && -1[-2w_1 + 0w_2 + 0w_3 + \cdots + 0w_{10} + b] \geq 1 && \iff && 2w_1 - b \geq 1, \\
&\text{B:} && -1[-w_1 + w_2 + 0w_3 + \cdots + 0w_{10} + b] \geq 1 && \iff && w_1 - w_2 - b \geq 1, \\
&\text{C:} && +1[w_1 + w_2 + 0w_3 + \cdots + 0w_{10} + b] \geq 1 && \iff && w_1 + w_2 + b \geq 1, \\
&\text{D:} && +1[2w_1 + 0w_2 + 0w_3 + \cdots + 0w_{10} + b] \geq 1 && \iff && 2w_1 + b \geq 1, \\
&\text{E:} && +1[2w_1 + 2w_2 + 0w_3 + \cdots + 0w_{10} + b] \geq 1 && \iff && 2w_1 + 2w_2 + b \geq 1.
\end{aligned} \tag{4.19}$$

This is not an unconstrained optimization problem. If one differentiates only the objective and sets its gradient to zero, then

$$\nabla_w \left(\frac{1}{2} \|w\|_2^2 \right) = w = 0. \tag{4.20}$$

At $w = 0$, the positive-class constraints require $b \geq 1$, whereas the negative-class constraints require $b \leq -1$; no intercept can satisfy both. The unconstrained minimum is therefore infeasible, which is precisely why the margin constraints must enter the optimality analysis through the Lagrangian and KKT conditions.

Solving the primal step by step. Add the simplified constraints for B and C:

$$(w_1 - w_2 - b) + (w_1 + w_2 + b) \geq 2 \quad \Longrightarrow \quad w_1 \geq 1. \tag{4.21}$$

It follows that every feasible point satisfies

$$\frac{1}{2} \|w\|_2^2 = \frac{1}{2} (w_1^2 + w_2^2 + \cdots + w_{10}^2) \geq \frac{1}{2}. \tag{4.22}$$

Choose $w_1 = 1.00$, $w_2 = \cdots = w_{10} = 0.00$. The B constraint then gives $1.00 - b \geq 1.00$, hence $b \leq 0.00$, while the C constraint gives $1.00 + b \geq 1.00$, hence $b \geq 0.00$. Thus $b = 0.00$. This candidate also satisfies the A, D, and E constraints, so it attains the lower bound and is globally optimal:

$$w^* = (1.00, 0.00, \ldots, 0.00)^\top, \qquad b^* = 0.00, \qquad p^* = \frac{1}{2} \|w^*\|_2^2 = 0.50. \tag{4.23}$$

The decision equation $(w^*)^\top x + b^* = 0$ reduces to $x_1 = 0.00$.

Table 4: Signed functional margins at the primal optimum.

Point	y_i	$(w^*)^\top x_i + b^*$	$y_i((w^*)^\top x_i + b^*)$	Status
A	-1	-2.00	2.00	Outside margin
B	-1	-1.00	1.00	Support vector
C	$+1$	1.00	1.00	Support vector
D	$+1$	2.00	2.00	Outside margin
E	$+1$	2.00	2.00	Outside margin

Table 4 confirms that B and C are support vectors because their signed margins equal 1.00. A, D, and E have signed margins strictly greater than 1.00 and therefore lie outside the margin strip.

Lagrangian, stationarity, and complementary slackness. Introduce one multiplier $\alpha_i \geq 0$ for each of the five constraints:

$$\mathcal{L}(w, b, \alpha) = \frac{1}{2} \|w\|_2^2 - \sum_{i \in \{A,B,C,D,E\}} \alpha_i \left[y_i (w^\top x_i + b) - 1 \right]. \tag{4.24}$$

Stationarity with respect to the primal variables gives

$$
\begin{aligned}
\nabla_w \mathcal{L} = 0 \quad &\Longrightarrow \quad w = \sum_i \alpha_i y_i x_i, \\
\frac{\partial \mathcal{L}}{\partial b} = 0 \quad &\Longrightarrow \quad \sum_i \alpha_i y_i = 0.
\end{aligned}
\tag{4.25}
$$

Complementary slackness requires

$$
\alpha_i \Big[y_i(w^\top x_i + b) - 1 \Big] = 0 \qquad \text{for every } i. \tag{4.26}
$$

For A, D, and E the bracket equals $2.00 - 1.00 = 1.00 > 0$, so complementary slackness forces

$$
\alpha_A = \alpha_D = \alpha_E = 0.00. \tag{4.27}
$$

For B and C the bracket is zero, so their multipliers may be positive. The intercept stationarity equation becomes

$$
-\alpha_B + \alpha_C = 0 \quad \Longrightarrow \quad \alpha_B = \alpha_C = \gamma. \tag{4.28}
$$

Substituting this equality into the weight stationarity equation gives

$$
\begin{aligned}
w &= \gamma(-1)(-1, 1, 0, \dots, 0) + \gamma(+1)(1, 1, 0, \dots, 0) \\
&= (2\gamma, 0, \dots, 0).
\end{aligned}
\tag{4.29}
$$

Matching the primal solution $w^* = (1.00, 0.00, \dots, 0.00)$ gives $2.00\gamma = 1.00$, and therefore

$$
\alpha_B = \alpha_C = 0.50. \tag{4.30}
$$

In the two visible coordinates, the reconstruction is

$$
0.50(-1)(-1.00, 1.00) + 0.50(+1)(1.00, 1.00) = (1.00, 0.00), \tag{4.31}
$$

which verifies that the two support vectors determine the separating direction.

Dual problem and strong duality. Minimizing the Lagrangian over w and b using the stationarity equations produces the hard-margin dual

$$
\begin{aligned}
\max_{\alpha} \quad & q(\alpha) = \sum_i \alpha_i - \frac{1}{2} \left\| \sum_i \alpha_i y_i x_i \right\|_2^2 \\
\text{subject to} \quad & \alpha_i \geq 0, \qquad \sum_i \alpha_i y_i = 0.
\end{aligned}
\tag{4.32}
$$

For this dataset, the two nonzero coordinates of $\sum_i \alpha_i y_i x_i$ are

$$
\begin{aligned}
z_1 &= 2\alpha_A + \alpha_B + \alpha_C + 2\alpha_D + 2\alpha_E, \\
z_2 &= -\alpha_B + \alpha_C + 2\alpha_E,
\end{aligned}
\tag{4.33}
$$

so the complete dataset-specific dual is

$$
\begin{aligned}
\max_{\alpha \geq 0} \quad & \alpha_A + \alpha_B + \alpha_C + \alpha_D + \alpha_E \\
& \qquad - \frac{1}{2} \Big[(2\alpha_A + \alpha_B + \alpha_C + 2\alpha_D + 2\alpha_E)^2 + (-\alpha_B + \alpha_C + 2\alpha_E)^2 \Big] \\
\text{subject to} \quad & -\alpha_A - \alpha_B + \alpha_C + \alpha_D + \alpha_E = 0.
\end{aligned}
\tag{4.34}
$$

At $(\alpha_A, \alpha_B, \alpha_C, \alpha_D, \alpha_E) = (0.00, 0.50, 0.50, 0.00, 0.00)$, the equality constraint holds and

$$q(\alpha) = 0.50 + 0.50 - \frac{1}{2}[(0.50 + 0.50)^2 + (-0.50 + 0.50)^2] = 0.50. \tag{4.35}$$

The primal construction already supplies a feasible value $p^* = 0.50$, while this dual-feasible point supplies the lower bound $d^* = 0.50$. Weak duality prevents a dual value from exceeding a primal value, so equality certifies that both points are optimal:

$$\boxed{d^* = 0.50 = p^*}. \tag{4.36}$$

This zero duality gap is the numerical manifestation of strong duality. The figure and all calculations in this subsection are reproduced by `scripts/build_svm_primal_dual_toy_example.py`.

5 Linear Soft-Margin SVM

5.1 Why Slack Variables Are Needed

The hard-margin analysis, with the detailed certificate given in Section 6.1, establishes that no (w, b) can satisfy all 114 unit-margin inequalities simultaneously. The soft-margin model changes the assumption rather than attempting to optimize over an empty set. For each observation it introduces a slack variable $\xi_i \geq 0$ that measures the shortfall from the desired signed functional margin of one:

$$\xi_i = \max\{0, 1 - t_i(w^\top x_i + b)\}. \tag{5.1}$$

When $\xi_i = 0$, the observation lies on or outside its supporting margin. If $0 < \xi_i \leq 1$, it lies inside the margin but remains on the correct side of the decision boundary. If $\xi_i > 1$, its signed score is negative and it is misclassified. This interpretation directly connects the optimization variables to the geometry of overlapping UPA and BPA observations.

5.2 Soft-Margin Primal Problem

The linear soft-margin primal problem is [1, 6]

$$\begin{aligned} \min_{w,b,\xi} \quad & p(w, b, \xi) = \frac{1}{2}\|w\|_2^2 + C\sum_{i=1}^{m} \xi_i \\ \text{subject to} \quad & t_i(w^\top x_i + b) \geq 1 - \xi_i, \qquad i = 1, \ldots, m, \\ & \xi_i \geq 0, \qquad i = 1, \ldots, m. \end{aligned} \tag{5.2}$$

The first term favors a wide margin, while the second penalizes total margin violation. The constant $C > 0$ controls their trade-off. Unlike the hard-margin problem, this primal is always feasible: $w = 0$, $b = 0$, and $\xi_i = 1$ satisfy every constraint. Eliminating the slack variables gives the equivalent unconstrained hinge-loss form

$$\min_{w,b} \quad \frac{1}{2}\|w\|_2^2 + C\sum_{i=1}^{m} \max\{0, 1 - t_i(w^\top x_i + b)\}. \tag{5.3}$$

The constrained form is retained below because it makes the primal–dual relationship and the role of C explicit.

5.3 Lagrangian and KKT Conditions

Introduce $\alpha_i \geq 0$ for $1 - \xi_i - t_i(w^\top x_i + b) \leq 0$ and $\mu_i \geq 0$ for $-\xi_i \leq 0$. The Lagrangian is

$$\begin{aligned} \mathcal{L}(w, b, \xi, \alpha, \mu) = \frac{1}{2}\|w\|_2^2 + C\sum_i \xi_i \\ + \sum_i \alpha_i[1 - \xi_i - t_i(w^\top x_i + b)] - \sum_i \mu_i\xi_i. \end{aligned} \tag{5.4}$$

Stationarity with respect to the primal variables gives

$$w = \sum_i \alpha_i t_i x_i, \qquad \sum_i \alpha_i t_i = 0, \qquad C - \alpha_i - \mu_i = 0. \tag{5.5}$$

Because $\mu_i \geq 0$, the final equality produces the upper bound $0 \leq \alpha_i \leq C$. The complete KKT system additionally requires primal feasibility, dual feasibility, and complementary slackness:

$$\begin{aligned} & t_i(w^\top x_i + b) - 1 + \xi_i \geq 0, \qquad \xi_i \geq 0, \\ & 0 \leq \alpha_i \leq C, \\ & \alpha_i[1 - \xi_i - t_i(w^\top x_i + b)] = 0, \\ & (C - \alpha_i)\xi_i = 0. \end{aligned} \tag{5.6}$$

These conditions are necessary and sufficient because the primal objective is convex, its constraints are affine, and strict feasibility is obtained by choosing sufficiently large positive slacks. They also explain the support vectors. An observation with $\alpha_i = 0$ does not determine w and lies outside the margin. A point with $0 < \alpha_i < C$ lies exactly on a supporting margin. A point with $\alpha_i = C$ may lie inside the margin or be misclassified, so it receives the largest multiplier permitted by the chosen penalty.

5.4 Soft-Margin Dual Problem

Minimizing the Lagrangian over w, b, and ξ using the stationarity conditions yields

$$\begin{aligned} \max_{\alpha} \quad & d(\alpha) = \sum_{i=1}^{m} \alpha_i - \frac{1}{2}\sum_{i=1}^{m}\sum_{j=1}^{m} \alpha_i\alpha_j t_i t_j x_i^\top x_j \\ \text{subject to} \quad & 0 \leq \alpha_i \leq C, \qquad \sum_{i=1}^{m} \alpha_i t_i = 0. \end{aligned} \tag{5.7}$$

The objective has the same form as the hard-margin dual, but the new box constraint $\alpha_i \leq C$ is essential. It prevents any single observation from exerting unlimited pressure on the separating hyperplane. Since the soft-margin primal is strictly feasible, strong duality holds and the optimal primal and dual values are equal up to numerical solver tolerance.

5.5 Selecting the Penalty Parameter by Grid Search

A small C places greater emphasis on a wide, strongly regularized margin and accepts more slack. A large C penalizes violations more severely and moves toward hard-margin behavior, which can increase sensitivity to overlapping observations. Because neither extreme is known in advance, C is selected from the half-logarithmic grid

$$\mathcal{C} = \{10^{k/2} : k = -4, -3, \ldots, 4\}. \tag{5.8}$$

Displayed to two decimal places, these values are

$$0.01,\ 0.03,\ 0.10,\ 0.32,\ 1.00,\ 3.16,\ 10.00,\ 31.62,\ 100.00. \tag{5.9}$$

Each candidate is evaluated by mean five-fold stratified validation AUROC. The log transformation, median imputation, standardization, and linear SVM are placed in one pipeline, so every preprocessing parameter is estimated from the corresponding training fold only.

Nested cross-validation separates model selection from performance estimation. In each of five outer folds, an independent five-fold inner grid search chooses C using only the outer training observations. The selected model then scores the untouched outer test fold. Table 5 shows that all five inner searches selected $C = 0.01$.

Table 5: Nested cross-validation for soft-margin penalty selection. AUROC values and C are displayed to two decimal places.

Outer fold	Train	Test	Selected C	Inner AUROC	Outer AUROC
1	91	23	0.01	0.95	0.92
2	91	23	0.01	0.96	0.95
3	91	23	0.01	0.94	0.98
4	91	23	0.01	0.91	0.98
5	92	22	0.01	0.95	0.89

5.6 Soft-Margin Solution on the Project Dataset

The final full-data grid search also selected $C = 0.01$, with mean validation AUROC 0.95. Figure 3 shows both the validation curve over C and the nested out-of-fold ROC curve. The nested out-of-fold AUROC is 0.95, with accuracy, sensitivity, and specificity all approximately 0.86 at decision threshold zero. Unlike the earlier squared-hinge result, these numbers are generated by the explicit linear soft-margin primal–dual model defined in this section.

The fitted standardized-space coefficients in Table 6 give the final linear scoring function. Their signs describe the direction of the score after the stated preprocessing but should not be interpreted as causal clinical effects.

Table 6: Final standardized-space soft-margin coefficients at $C = 0.01$.

Parameter	Estimate	Parameter	Estimate
PAC	0.29	18-oxoF	0.22
PRA	-0.18	Systolic BP	0.09
Potassium	-0.19	Diastolic BP	0.18
Tumor size	0.15	DDD	0.07
18-OHF	0.23	Age	-0.08
Intercept	-0.02		

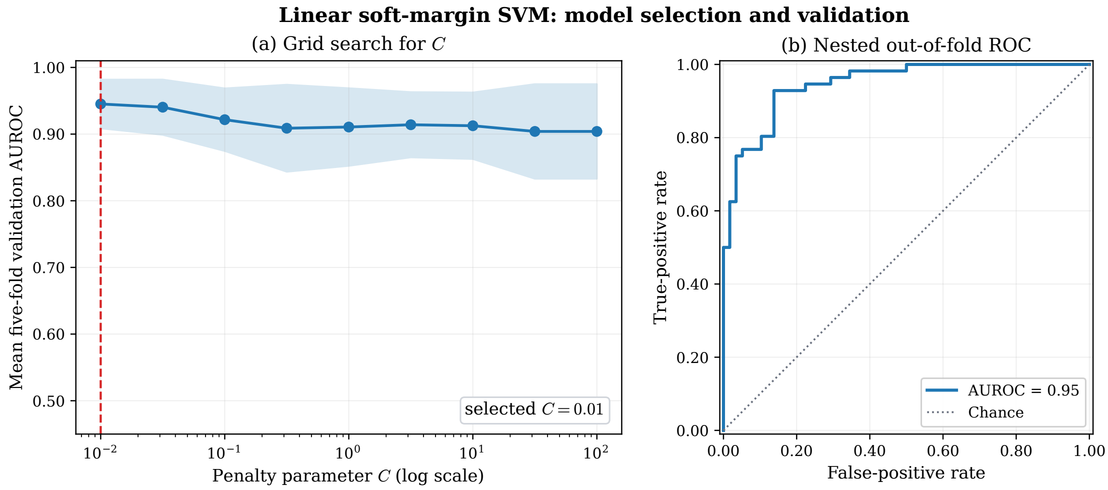

Figure 3: Soft-margin model selection and validation. Panel (a) reports mean five-fold validation AUROC with one standard deviation across folds for every candidate C; the dashed line marks the selected value. Panel (b) reports the nested out-of-fold ROC obtained when C is selected inside each outer training fold.

The full-data solution contains 82 support vectors and 77 observations with positive slack, with $\sum_i \xi_i = 46.89$. Substitution into the primal gives

$$\begin{aligned} p^* &= \frac{1}{2}\|w\|_2^2 + C\sum_i \xi_i \\ &= \frac{1}{2}(0.33) + (0.01)(46.89) \\ &\approx 0.63. \end{aligned} \tag{5.10}$$

The dual calculation gives $d^* \approx 0.63$, and the full-precision primal–dual gap is 4.85×10^{-10}. The remaining numerical checks are $|\sum_i \alpha_i t_i| = 2.78 \times 10^{-17}$ and a zero weight-reconstruction residual at stored precision. The two maximum complementary-slackness residuals are 2.35×10^{-10} and 8.57×10^{-11}, confirming the KKT conditions to solver tolerance. Table 7 summarizes the model-selection and optimization results.

Table 7: Soft-margin model-selection, validation, and optimization summary.

Quantity	Value
Selected C from full-data grid search	0.01
Mean inner-CV AUROC at selected C	0.95
Nested out-of-fold AUROC	0.95
Nested out-of-fold accuracy	0.86
Nested sensitivity	0.86
Nested specificity	0.86
Full-data support vectors	82
Full-data observations with $\xi_i > 0$	77
Full-data $\sum_i \xi_i$	46.89
Full-data primal objective	0.63
Full-data dual objective	0.63

All full-precision grid-search results, nested predictions, coefficients, support-vector multipliers, slacks, and primal–dual checks are written under `output/csv/` by `scripts/linear_soft_` `margin_svm.py`. Tables in this section and both panels of Figure 3 are generated from those Python outputs.

The penalty C is selected by the grid search of Figure 4: validation AUROC is highest at the strongly regularised end and the training–validation gap is smallest there, so $C = 0.01$ is preferred. Figure 5 recasts the same grid as a Pareto trade-off between validation performance and overfitting, on which $C = 0.01$ is the single non-dominated point. Figure 6 shows the resulting soft-margin boundary in the first two principal components: the separating line passes straight through the region where the two subtypes overlap, accepting the misclassifications it cannot avoid and charging for them through the slack variables.

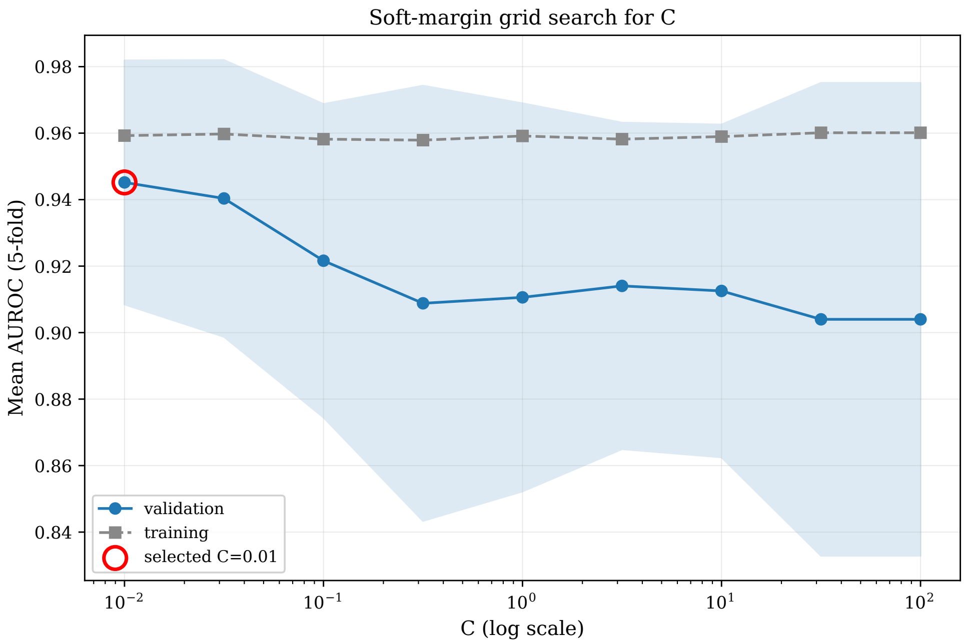

Figure 4: Grid search for the penalty C. Mean five-fold validation AUROC (with training AUROC for reference) across the C grid; the selected value $C = 0.01$ is circled. Stronger regularisation generalises best on this overlapping data.

5.7 Solving the Primal Directly: the Stochastic Gradient Method

The dual quadratic program above is solved in this report by `scikit-learn`'s SMO routine. The soft-margin *primal* can instead be minimized directly, and because it is a regularized finite sum over the m training points it is a natural target for the stochastic gradient method (SG). Writing the homogeneous primal in the finite-sum form

$$\min_{w\in\mathbb{R}^p} \frac{1}{m}\sum_{i=1}^{m} \max(0,\, 1 - t_i\, w^\top x_i) + \frac{\lambda}{2}\|w\|^2, \tag{5.11}$$

each step samples one index k uniformly at random and takes a subgradient step. The hinge term is non-differentiable at its kink, so the lecture uses its subgradient: $0 \in \partial\ell$ whenever $t_k\, w^\top x_k \geq 1$, and $\partial\ell = -t_k x_k$ otherwise. Adding the subgradient of the $\frac{\lambda}{2}\|w\|^2$ regularizer and taking the Pegasos step size $\eta_k = 1/(\lambda k)$ gives the update

$$w_{k+1} = w_k - \eta_k\, g_k, \qquad g_k = \begin{cases} \lambda w_k, & t_k\, w_k^\top x_k \geq 1, \\ \lambda w_k - t_k x_k, & t_k\, w_k^\top x_k < 1. \end{cases} \tag{5.12}$$

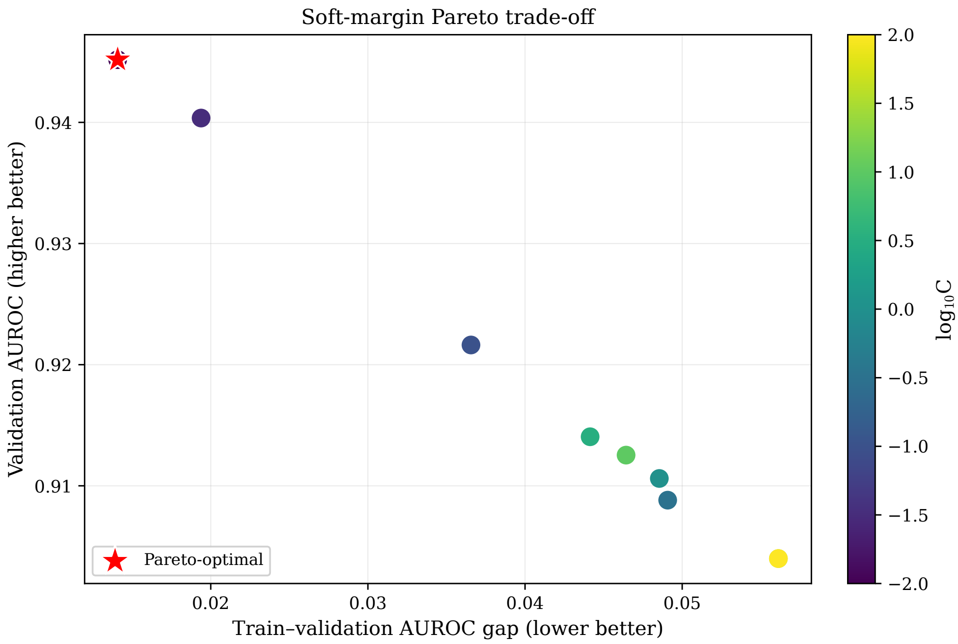

Figure 5: The same grid as a Pareto trade-off between mean validation AUROC (higher is better) and the train–validation AUROC gap (lower is better). Points are coloured by $\log_{10} C$; $C = 0.01$ (star) is the single Pareto-optimal setting.

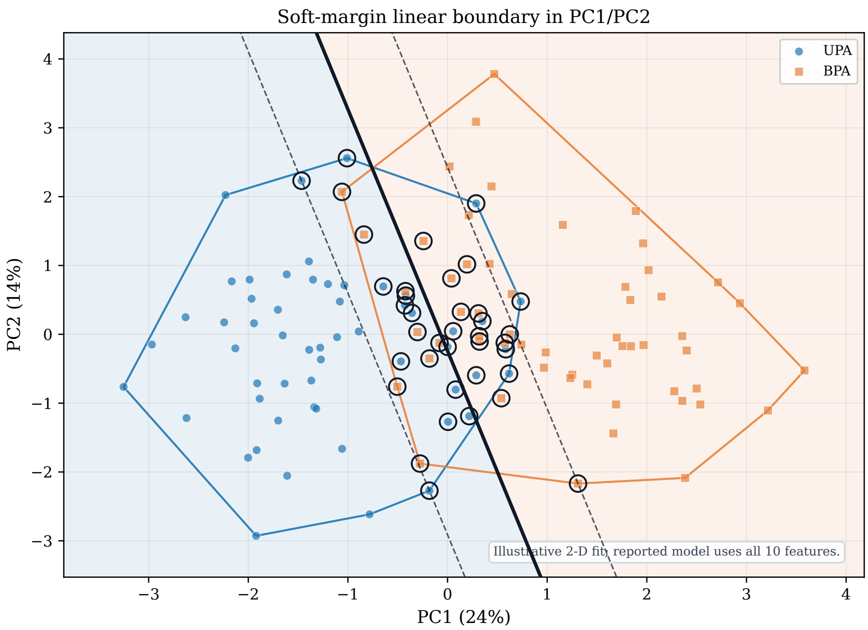

Figure 6: Fitted soft-margin decision boundary in the first two principal components, with the ± 1 margins and the circled support vectors. The boundary is an illustrative two-dimensional fit; the reported model is trained on all ten features.

The $\frac{\lambda}{2}\|w\|^2$ term makes the objective λ-strongly convex [9], and the intercept is recovered by appending a constant feature to each x_i; this is the stochastic gradient method of the course lecture [10] applied to the regularised hinge objective. Figure 7 shows the run on the standardized project data ($m = 114$, $\lambda = 0.01$). The reference optimum $P^\star = 0.250$ is obtained from a primal hinge solver; the SG direction agrees with that solution to a cosine similarity of 0.99 and reproduces its discrimination (AUROC 0.958 versus 0.955), confirming that the stochastic primal route reaches the same separating hyperplane as the dual solver.

Table 8: Steps for the stochastic gradient method to reach a given optimality gap $P(w_k) - P^\star$ on the project data, measured on the best-so-far iterate. One epoch is a full pass over the $m = 114$ points.

Target optimality gap	Steps	Epochs
Within 5% of $P^\star$	1,653	≈ 14
Within 1% of $P^\star$	7,591	≈ 67

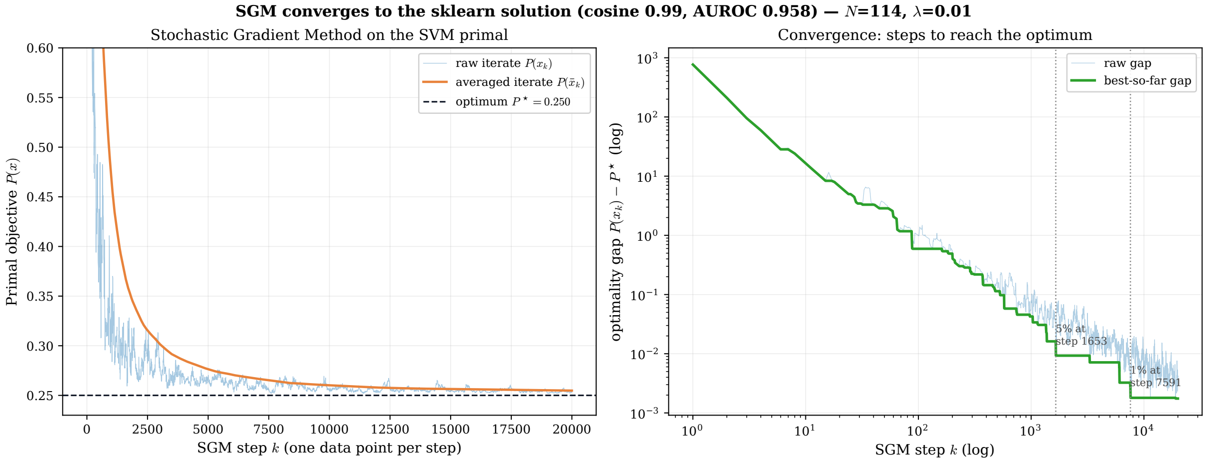

Figure 7: Stochastic gradient method on the soft-margin primal. Left: the raw iterate $P(w_k)$ (thin) fluctuates around the optimum $P^\star$ because each step uses a single random point, while the running (Polyak) average $P(\bar{w}_k)$ converges smoothly. Right: the optimality gap $P(w_k) - P^\star$ on log–log axes, with the best-so-far gap crossing 5% of $P^\star$ at about step 1,650 (≈ 14 epochs) and 1% at about step 7,600 (≈ 67 epochs).

The convergence behaviour is worth reading carefully. For a strongly convex objective the stochastic gradient analysis gives a clean linear rate only in the interpolation regime, where a single w drives every per-point loss to zero so that the gradient-noise floor B vanishes. The project data are *not* linearly separable—this is precisely what the infeasibility certificate of the next section establishes—so the hinge loss cannot be driven to zero, the noise floor is strictly positive ($B > 0$), and the iterate settles into a persistent band around the optimum rather than converging geometrically. The averaged iterate is what supplies a stable solution, and the observed sublinear, noisy decay in Figure 7 is the expected signature of this $B > 0$ regime.

6 Detailed Hard-Margin Numerical Certificate

6.1 Full-Data Infeasibility Certificate

Three mutually reinforcing checks are used on the complete processed dataset, in the order direct solver status, algebraic Farkas certificate, and geometric convex-hull interpretation. Here

$U \in \mathbb{R}^{114\times 11}$ contains the ten standardized features and the intercept column.

Homework-style direct solver verification. The primal model supplied to a constrained solver is

$$\begin{aligned} \min_{w,b} \quad & \frac{1}{2}\|w\|_2^2 \\ \text{subject to} \quad & U\theta \geq \mathbf{1}. \end{aligned} \tag{6.1}$$

Before an objective can be minimized, the solver must find a point satisfying all 114 constraints. The implementation therefore first solves the equivalent feasibility model $\min_\theta 0$ subject to $-U\theta \leq -\mathbf{1}$; changing the objective cannot make an empty feasible set nonempty. The direct solver audit returned the result in Table 9.

Table 9: Direct full-data primal feasibility check.

Item	Solver result
Decision variables $(w_1, \ldots, w_{10}, b)$	11
Hard-margin constraints	114
Solver	SciPy HiGHS
HiGHS model status	Status 8: Infeasible
Primal parameters (w, b) returned	None
Primal objective $\frac{1}{2}\|w\|_2^2$	Not defined

Thus the correct statement is that the solver returns an *infeasible status*; it does not return an "infeasible solution." No primal solution exists for the solver to report. This is the direct computational verification used in the homework-style workflow.

Independent certificate. The solver status is then strengthened by a direct, independently checkable proof. Farkas' lemma states that the system $U\theta \geq \mathbf{1}$ is infeasible if there is a vector $\lambda \in \mathbb{R}^{114}$ satisfying

$$\lambda \geq 0, \qquad U^\top \lambda = 0, \qquad \mathbf{1}^\top \lambda = 1. \tag{6.2}$$

The auxiliary linear program for this certificate returned twelve nonzero weights, shown in Table 10; the other 102 weights are zero. Displayed weights use two decimal places as requested. Every check uses the unrounded values in the companion CSV file, which also contains a separate two-decimal display column; the displayed values alone should not be used to recompute the certificate.

Table 10: Nonzero weights in the full-data Farkas infeasibility certificate.

Patient	Class	λ_i	Patient	Class	λ_i
CGH-021	UPA	0.21	CGH-072	BPA	0.04
CGH-024	BPA	0.05	CGH-081	UPA	0.01
CGH-026	BPA	0.16	CGH-092	BPA	0.01
CGH-053	UPA	0.04	CGH-105	UPA	0.01
CGH-058	UPA	0.17	CGH-108	BPA	0.05
CGH-060	BPA	0.18	CGH-111	UPA	0.07

The full-precision numerical audit gives

$$
\begin{aligned}
\mathbf{1}^\top \lambda &= 1.00, \\
\sum_{i:y_i=+1} \lambda_i &= 0.50, \\
\sum_{i:y_i=-1} \lambda_i &= 0.50, \\
\|U^\top \lambda\|_\infty &= 2.50 \times 10^{-16}.
\end{aligned}
\tag{6.3}
$$

The first three quantities are displayed to two decimal places. The final residual is retained in scientific notation because rounding it to 0.00 would hide the numerical accuracy of the certificate. Now suppose, for contradiction, that a hard-margin parameter θ satisfied $U\theta \geq \mathbf{1}$. Nonnegativity of λ would imply

$$
\underbrace{(U^\top \lambda)^\top}_{0} \theta = \lambda^\top U\theta \geq \lambda^\top \mathbf{1} = 1, \tag{6.4}
$$

which is the contradiction $0 \geq 1$. Therefore the complete hard-margin constraint system has no feasible solution. In fact, the twelve observations in Table 10 already provide a sufficient infeasible subsystem.

The same certificate has a geometric interpretation. Because the intercept component of $U^\top \lambda = 0$ gives equal class masses of 0.5, define $\beta_i^+ = 2\lambda_i$ for UPA and $\beta_i^- = 2\lambda_i$ for BPA. Each set of coefficients is nonnegative and sums to one, while the feature components of $U^\top \lambda = 0$ give

$$
\sum_{i:y_i=+1} \beta_i^+ x_i = \sum_{i:y_i=-1} \beta_i^- x_i. \tag{6.5}
$$

Thus the standardized UPA and BPA convex hulls contain the same point. Their largest coordinate difference is only 6.66×10^{-16}, as illustrated in Figure 8. Intersecting convex hulls cannot be strictly separated by a hyperplane.

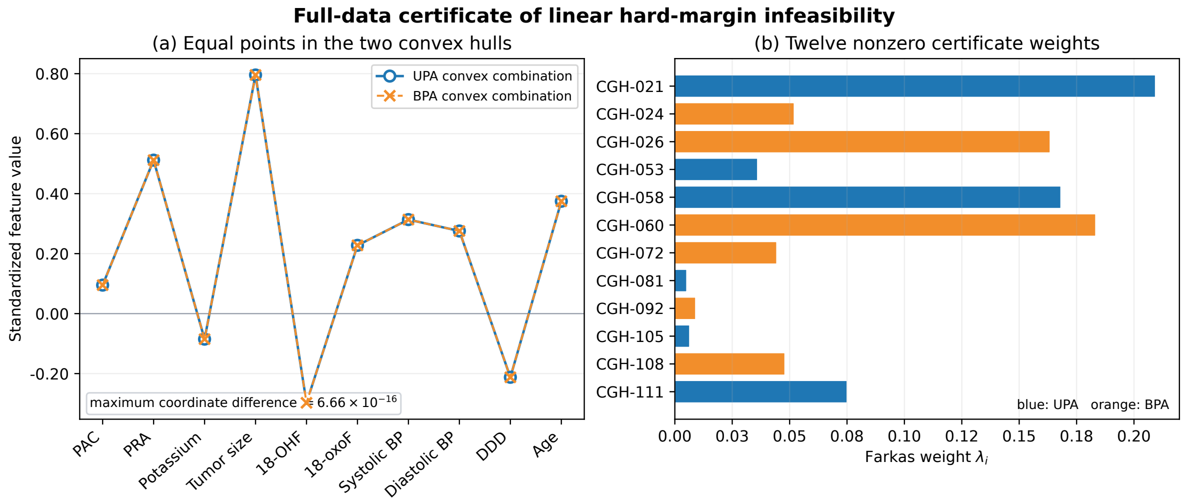

Figure 8: Full-data infeasibility certificate. Panel (a) shows the identical ten-dimensional point obtained as a convex combination of UPA observations and as a convex combination of BPA observations; the plot compares coordinates directly and is not a two-dimensional projection. Panel (b) shows the twelve nonzero Farkas weights.

Two complementary illustrations make the certificate intuitive. Figure 9 contrasts the two systems of Farkas' lemma: either a separating hyperplane exists (System A), or a nonnegative combination of the constraints proves a contradiction $0 \geq 1$ (System B), never both. Figure 10 shows the same fact geometrically on the project data projected to its first two principal components: the convex hulls of the two subtypes overlap, and overlapping hulls cannot be separated by any hyperplane. The theory-of-the-alternative structure behind this argument follows the treatment of duality and first-order optimality in the course lectures [7, 8].

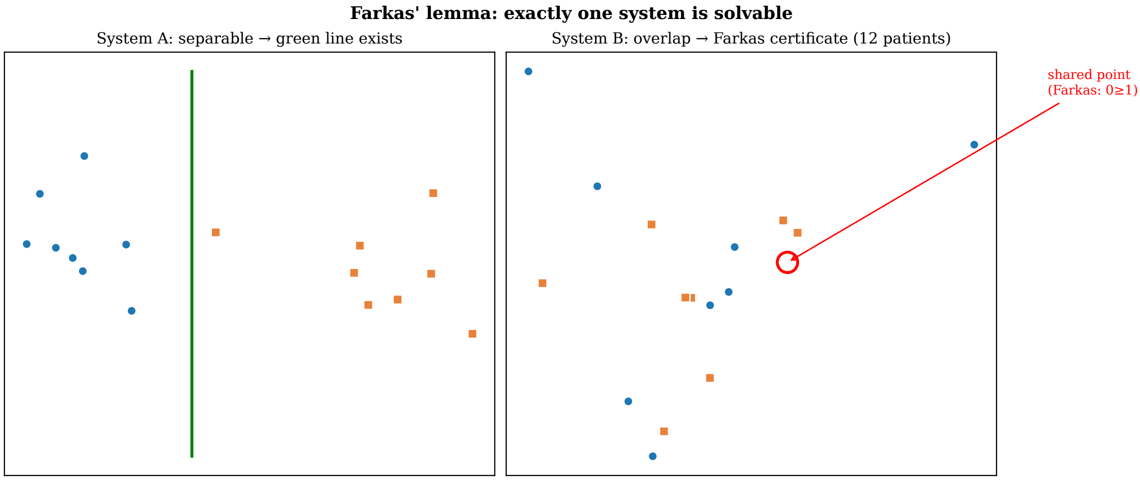

Figure 9: Farkas' lemma as a theorem of the alternative. System A (left): the classes are separable, so a hyperplane exists. System B (right): the classes overlap, so a nonnegative weighting of the points certifies infeasibility ($0 \geq 1$). Exactly one system is solvable; for the project data it is System B.

Consequently, for the current preprocessed project data,

$$\boxed{\text{the UPA and BPA classes are not linearly separable.}} \tag{6.6}$$

The certificate, CSV output, numerical checks, and figure are reproduced by `scripts/build_full_data_infeasibility_certificate.py`. The script writes four machine-readable audit files under `output/csv/`: the direct solver status, sparse Farkas vector, two convex-hull representations, and numerical residual checks. The two tables above are generated from the same Python results and then included in this PDF.

6.2 Supplementary Ten-Observation Spreadsheet Illustration

To make the matrix constraint concrete, a self-contained teaching table was constructed from the first ten observations in the project spreadsheet. The selection rule is transparent: scan from top to bottom, take the first five UPA and first five BPA observations, and restore source-row order. In this dataset these are exactly rows 2–11. The four hormone variables are log-transformed and all ten features are standardized using statistics from these ten rows. No selected value is missing, so median imputation is inactive.

For $p = 10$, the vectors used in one constraint row are

$$\theta = \begin{bmatrix} w_1 & \cdots & w_{10} & b \end{bmatrix}^\top, \qquad u_i = t_i \begin{bmatrix} x_{i1} & \cdots & x_{i,10} & 1 \end{bmatrix}^\top. \tag{6.7}$$

Their inner product is the signed functional margin:

$$u_i^\top \theta = t_i(x_i^\top w + b) = t_i s_i. \tag{6.8}$$

The expression is therefore not "$u_i > 1$": u_i is a vector. The scalar dot product $u_i^\top \theta$ must satisfy $u_i^\top \theta \geq 1$. For UPA, $t_i = +1$ and the rule becomes $s_i \geq 1$; for BPA, $t_i = -1$ and it becomes $s_i \leq -1$.

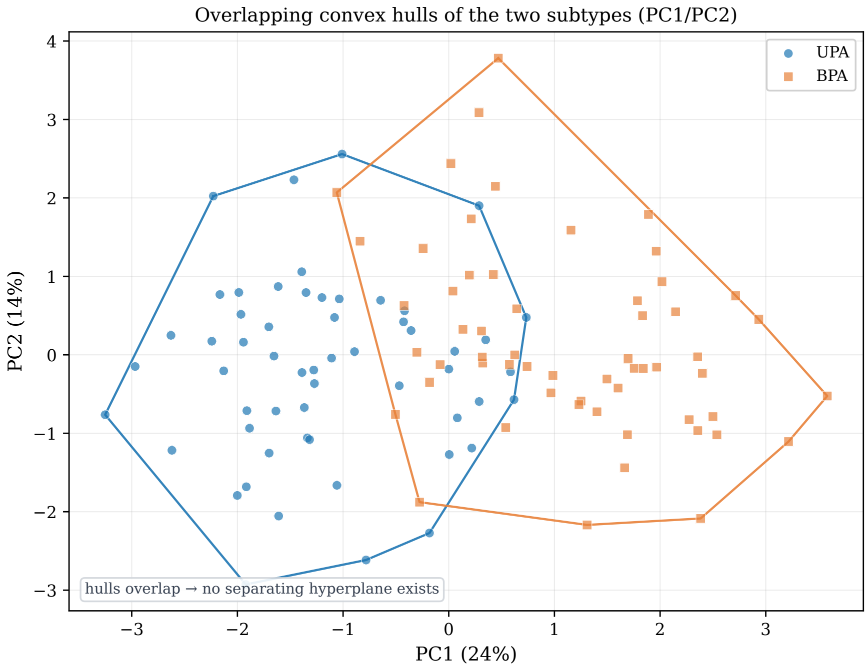

Figure 10: Convex hulls of the UPA and BPA subtypes in the first two principal components of the standardised data. The hulls overlap, so no separating hyperplane exists—the geometric statement of the infeasibility certificate. The projection is for visualisation; the certificate itself is computed in the full ten-dimensional space.

Solving the primal quadratic program for this teaching subsystem gives

$$\begin{aligned} w_{\text{teach}}^{\top} &\approx (0.34, -0.42, -0.65, 0.08, -0.10, \\ &\qquad 0.16, -0.13, 0.39, -0.75, 0.64), \\ b_{\text{teach}} &\approx 0.29. \end{aligned} \tag{6.9}$$

These coefficients are displayed to two decimal places; all constraint and objective checks use the full-precision values stored in the generated CSV files.

Numerical evaluation of the objective. The ten observations contribute ten margin constraints; they do not create ten additive loss terms. Thus the fitted teaching subsystem has one global primal objective. Substituting the fitted weights gives

$$\begin{aligned} \|w_{\text{teach}}\|_2^2 &\approx (0.34)^2 + (-0.42)^2 + (-0.65)^2 + (0.08)^2 + (-0.10)^2 \\ &\quad + (0.16)^2 + (-0.13)^2 + (0.39)^2 + (-0.75)^2 + (0.64)^2 \\ &\approx 1.90 \quad \text{from displayed coefficients,} \\ &= 1.88 \quad \text{from full-precision coefficients,} \\ \phi(\theta_{\text{teach}}) &= \frac{1}{2}\|w_{\text{teach}}\|_2^2 \\ &= \frac{1}{2}(1.88) \\ &\approx \boxed{0.94}. \end{aligned} \tag{6.10}$$

The displayed arithmetic is rounded to two decimal places, while the reported feasibility status uses the full-precision solver coefficients. The intercept $b_{\text{teach}} \approx 0.29$ is not squared because the hard-margin primal does not penalize the intercept. The ten rowwise quantities governed by this solution are the constraints $u_i^{\top}\theta \geq 1$ in Table 11.

The coefficient order is PAC, PRA, potassium, tumor size, 18-OHF, 18-oxoF, systolic blood pressure, diastolic blood pressure, DDD, and age. Table 11 shows every constraint. The complete raw x_i, transformed x_i, standardized x_i, u_i, parameter, gradient, and Hessian tables are supplied in the companion CSV folder `output/csv/hard_margin_10row/`.

Table 11: Teaching-only row-by-row evaluation of $u_i^\top \theta \geq 1$. Displayed values are rounded; feasibility uses unrounded values.

Patient	Class	t_i	Score s_i	$u_i^\top \theta$	$g_i = 1 - u_i^\top \theta$	Feasible	SV
CGH-001	UPA	$+1$	3.22	3.22	-2.22	Yes	No
CGH-002	UPA	$+1$	1.39	1.39	-0.39	Yes	No
CGH-003	UPA	$+1$	1.00	1.00	0.00	Yes	Yes
CGH-004	BPA	-1	-1.00	1.00	0.00	Yes	Yes
CGH-005	BPA	-1	-1.41	1.41	-0.41	Yes	No
CGH-006	BPA	-1	-1.59	1.59	-0.59	Yes	No
CGH-007	BPA	-1	-1.00	1.00	0.00	Yes	Yes
CGH-008	BPA	-1	-1.00	1.00	0.00	Yes	Yes
CGH-009	UPA	$+1$	2.27	2.27	-1.27	Yes	No
CGH-010	UPA	$+1$	1.00	1.00	0.00	Yes	Yes

For example, CGH-004 is a BPA observation, so $t_4 = -1$. Its score is $s_4 = -1$, giving $u_4^\top \theta = (-1)(-1) = 1$ and $g_4 = 0$. It lies exactly on the negative supporting margin and is a support vector. The corresponding spreadsheet calculation is

$$\begin{aligned} s_i &= \mathrm{SUMPRODUCT}(x_i, w) + b, \\ u_i^\top \theta &= t_i s_i, \\ \text{constraint check} &= (u_i^\top \theta \geq 1 - 10^{-8}). \end{aligned} \tag{6.11}$$

Figure 11 plots the raw scores rather than the signed margins. This preserves the two class regions: BPA observations lie at or left of -1, UPA observations lie at or right of $+1$, and $s = 0$ is the classifier boundary.

This feasible subsystem does not imply that the complete cohort is separable. Removing 104 constraints can leave a feasible problem even when adding the remaining constraints makes the full system infeasible. No ROC, AUROC, sensitivity, or specificity is calculated from this ten-row in-sample illustration.

6.3 Cross-Validation and ROC Rule

The feasibility audit uses stratified five-fold cross-validation with shuffling and random seed zero. All preprocessing operations are fitted inside the training fold. In each fold, the hard-margin classifier is fitted only if the training constraints are feasible. Held-out decision scores would then be pooled to construct the ROC curve, and AUROC would summarize discrimination across score thresholds.

This ordering is important. ROC analysis requires a fitted scoring function for every held-out observation. If a training fold is infeasible, there are no valid hard-margin parameters for that fold and therefore no legitimate out-of-fold scores. Reporting an ROC curve in that situation would require silently changing either the model or the data.

The worked subsystem in Section 6.2 is not a cross-validation split and is not used to generate ROC results.

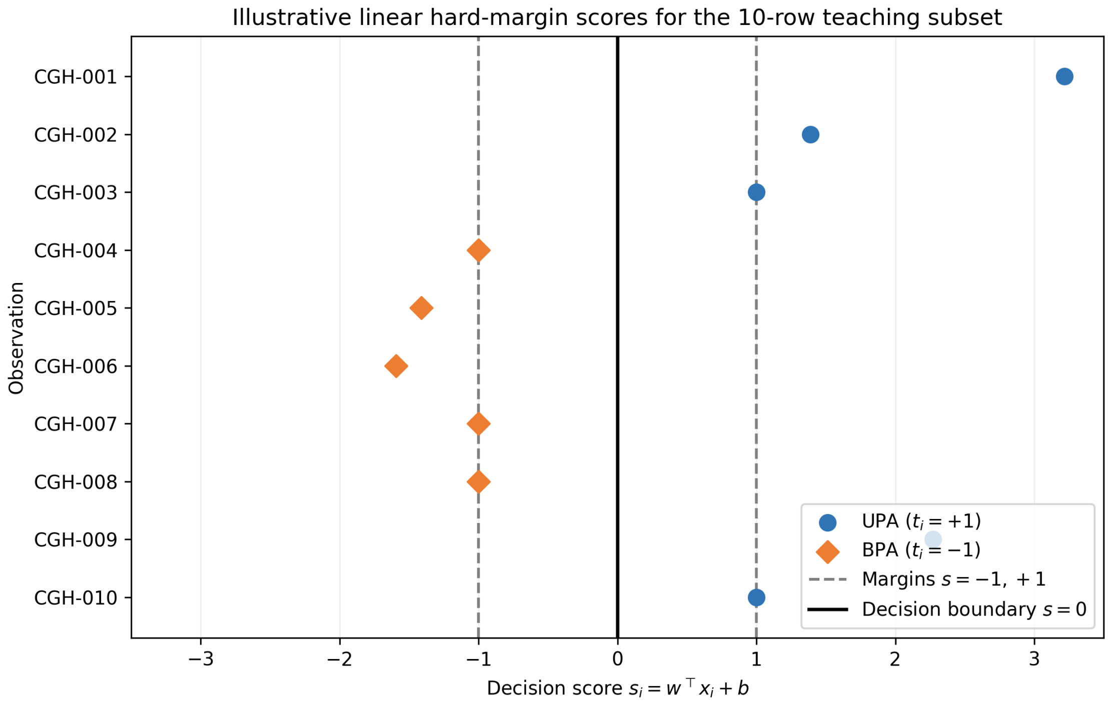

Figure 11: Decision scores for the separable ten-row teaching subsystem. This is an algebraic illustration, not a validation or ROC result.

7 Kernel Trick: RBF Soft-Margin SVM

7.1 From a Linear Boundary to a Kernel Boundary

The strong linear soft-margin result does not prove that a nonlinear boundary is unnecessary; it only establishes a competitive baseline. The kernel trick tests nonlinear structure without explicitly constructing a potentially high-dimensional feature vector. Let $\varphi(x)$ map an observation into a feature space $\mathcal{H}$. The soft-margin primal in that space is

$$\begin{aligned} \min_{w,b,\xi} \quad & \frac{1}{2}\|w\|_{\mathcal{H}}^2 + C\sum_{i=1}^{m}\xi_i \\ \text{subject to} \quad & t_i(\langle w, \varphi(x_i)\rangle_{\mathcal{H}} + b) \geq 1 - \xi_i, \qquad \xi_i \geq 0. \end{aligned} \tag{7.1}$$

The corresponding dual depends on transformed observations only through inner products:

$$\begin{aligned} \max_{\alpha} \quad & \sum_i \alpha_i - \frac{1}{2}\sum_i\sum_j \alpha_i\alpha_j t_i t_j K(x_i, x_j) \\ \text{subject to} \quad & 0 \leq \alpha_i \leq C, \qquad \sum_i \alpha_i t_i = 0, \end{aligned} \tag{7.2}$$

where $K(x_i, x_j) = \langle \varphi(x_i), \varphi(x_j)\rangle_{\mathcal{H}}$. The fitted score is

$$f(x) = \sum_{i\in\mathcal{S}} \alpha_i t_i K(x_i, x) + b, \tag{7.3}$$

so only the support-vector set $\mathcal{S}$ is needed at prediction time. Kernelization does not replace the soft margin: C still bounds each dual multiplier and controls the penalty for overlap.

This analysis uses the radial basis function (RBF) kernel

$$K(x_i, x_j) = \exp[-\gamma\|x_i - x_j\|_2^2]. \tag{7.4}$$

Small γ produces a smooth, slowly varying decision surface and tends toward linear-like behavior over a limited data region. Large γ makes each observation influential only in a narrow neighborhood, allowing a highly curved boundary that can memorize a small training cohort. Therefore C and γ must be tuned jointly.

7.2 Validity of the RBF Kernel: Positive Semi-Definiteness and Feature Map

Everything above presumes that some feature map φ exists with $K(x, y) = \langle\varphi(x), \varphi(y)\rangle_{\mathcal{H}}$. That is not automatic: an arbitrary similarity function need not be an inner product in any space, and if it is not, the dual loses concavity and the optimization guarantees of Section 4.5 no longer apply. This subsection establishes the two properties that license the RBF kernel.

A symmetric kernel K is *positive semi-definite* (PSD) if for every finite set $x_1, \ldots, x_m$ the Gram matrix $G_{ij} = K(x_i, x_j)$ satisfies $c^\top Gc \geq 0$ for all $c \in \mathbb{R}^m$. Four standard closure rules are used [2, 6]: if K_1, K_2 are PSD then (K1) $\lambda_1 K_1 + \lambda_2 K_2$ is PSD for $\lambda_1, \lambda_2 \geq 0$; (K2) the product $K_1 K_2$ is PSD, by the Schur product theorem; (K3) $K(x, y) = f(x)f(y)$ is PSD for any real f; and (K4) a pointwise limit of PSD kernels is PSD.

The RBF kernel is PSD. The linear kernel is PSD because

$$c^\top Gc = \sum_i \sum_j c_i c_j \, x_i^\top x_j = \Big\| \sum_i c_i x_i \Big\|_2^2 \geq 0, \tag{7.5}$$

so $(x^\top y)^k$ is PSD for every integer $k \geq 0$ by repeated application of (K2). The exponential series has non-negative coefficients, hence

$$\exp\!\left[2\gamma \, x^\top y\right] = \sum_{k=0}^{\infty} \frac{(2\gamma)^k}{k!} \, (x^\top y)^k \tag{7.6}$$

is a limit of non-negative combinations of PSD kernels and is PSD by (K1) and (K4). Expanding the squared norm as $\|x - y\|_2^2 = \|x\|_2^2 - 2x^\top y + \|y\|_2^2$ factors the RBF kernel into

$$\exp\!\left[-\gamma \|x - y\|_2^2\right] = \underbrace{e^{-\gamma\|x\|_2^2} e^{-\gamma\|y\|_2^2}}_{\text{PSD by (K3)}} \cdot \underbrace{\exp\!\left[2\gamma \, x^\top y\right]}_{\text{PSD}}, \tag{7.7}$$

which is PSD by (K2). The Gram matrix is therefore PSD for any dataset and any $\gamma > 0$.

An explicit feature map. The same expansion produces φ directly. Writing the multinomial expansion of $(x^\top y)^k$ over multi-indices $\alpha \in \mathbb{N}_0^p$, with $|\alpha| = \sum_j \alpha_j$, $\alpha! = \prod_j \alpha_j!$ and $x^\alpha = \prod_j x_j^{\alpha_j}$, gives

$$\exp\!\left[2\gamma \, x^\top y\right] = \sum_{\alpha \in \mathbb{N}_0^p} \frac{(2\gamma)^{|\alpha|}}{\alpha!} \, x^\alpha y^\alpha. \tag{7.8}$$

Reinstating the two Gaussian factors and splitting each term symmetrically yields $K(x, y) = \langle\varphi(x), \varphi(y)\rangle$ with components

$$\varphi_\alpha(x) = e^{-\gamma\|x\|_2^2} \sqrt{\frac{(2\gamma)^{|\alpha|}}{\alpha!}} \, x^\alpha, \qquad \alpha \in \mathbb{N}_0^p. \tag{7.9}$$

The index set is all multi-indices, so $\mathcal{H}$ is infinite-dimensional: φ contains every monomial in the p features, damped by $e^{-\gamma\|x\|_2^2}$ and weighted so the series converges.

Consequences. Three points follow. First, φ can never be formed explicitly, which is exactly why the kernel trick is required rather than merely convenient: the scalar $K(x_i, x_j)$ stands in for an inner product between infinite vectors. Second, PSD of the Gram matrix makes the dual objective concave, so the kernelized problem is a convex quadratic program with a unique optimum rather than a surface with spurious local maxima. Third, the primal in $\mathcal{H}$ is convex with affine constraints and admits the feasible point $(w, b, \xi) = (0, 0, \mathbf{1})$, so Slater's condition holds exactly as in Section 4.5; the primal–dual gap of 2.2×10^{-11} reported for the fitted RBF model is therefore a meaningful check that the solver reached that optimum.

7.3 Joint Grid Search and Nested Validation

The RBF analysis uses the same log transformation, median imputation, standardization, stratified folds, and random seed as the linear analysis. The joint grid contains nine values of C from 0.01 to 100.00 and nine values of γ from 1.00×10^{-3} to 10.00, both at half-logarithmic spacing. Thus each inner search evaluates 81 parameter pairs by mean AUROC.

In every outer fold, preprocessing and the complete 81-point grid search are fitted using only the outer training data. Table 12 shows that the selected pairs vary substantially across folds. This instability is itself relevant: with only 114 observations, the apparently best nonlinear scale depends on which observations enter the training set.

Table 12: Nested cross-validation for joint RBF tuning. AUROC values and parameter displays use two decimals or two-decimal scientific notation.

Fold	C	γ	Inner train	Inner validation	Outer AUROC
1	3.16	1.00e-03	0.97	0.95	0.92
2	10.00	1.00e-03	0.96	0.96	0.94
3	0.32	0.03	0.96	0.94	0.98
4	1.00	3.16e-03	0.95	0.92	0.98
5	0.01	1.00e-03	0.97	0.96	0.88

The full-data grid search selected $C = 0.01$ and $\gamma = 1.00 \times 10^{-3}$. This γ is the lower boundary of the prespecified grid, indicating that the validation criterion favors the smoothest candidate rather than a strongly nonlinear boundary. At the selected pair, mean training AUROC is 0.96 and mean validation AUROC is 0.95, a gap of only 0.01. The induced-feature-space primal and dual objectives both round to 1.12, with full-precision duality gap 2.23×10^{-11}.

Table 13: Full-data RBF model-selection and optimization audit.

Quantity	Value
Selected C	0.01
Selected γ	1.00e-03
Mean training AUROC	0.96
Mean validation AUROC	0.95
Train–validation gap	0.01
Largest gap anywhere on grid	0.46
Mean nested outer AUROC	0.94
Support vectors	113
Observations with $\xi_i > 0$	111
Primal objective	1.12
Dual objective	1.12

7.4 Overfitting Diagnostic

The selected RBF model is not strongly overfit, but the grid contains a clear demonstration of the risk. Figure 12 compares validation AUROC with the train–validation gap. At $C = 0.01$ and $\gamma = 10.00$, training AUROC is 1.00 while validation AUROC falls to 0.54, giving a gap of 0.46. Across all values of C at $\gamma = 10.00$, training AUROC remains 1.00 but validation AUROC is only 0.54–0.59. The high-γ kernel has therefore memorized the training folds without learning a stable decision rule.

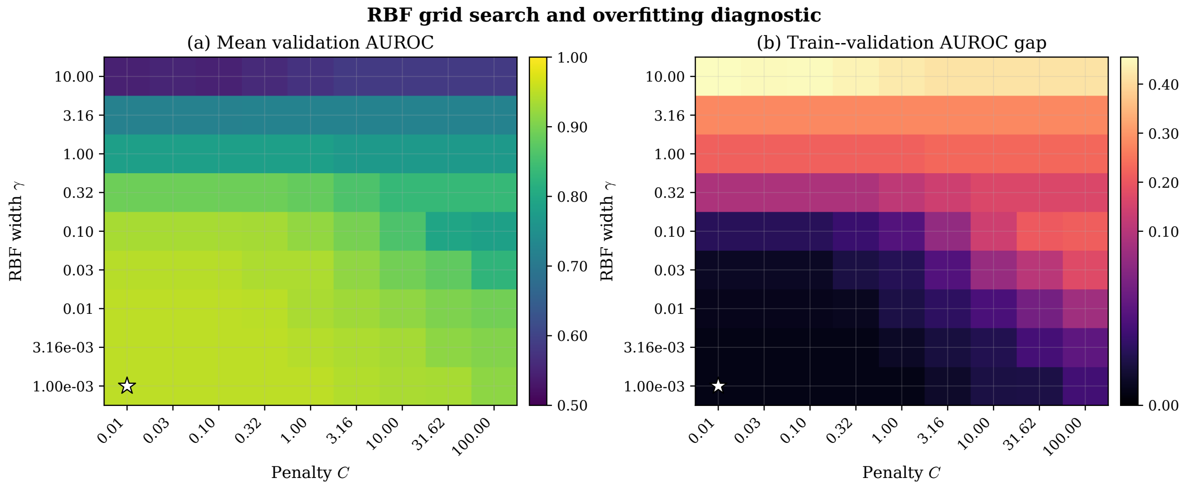

Figure 12: Joint RBF grid search. Panel (a) shows mean validation AUROC. Panel (b) shows mean training AUROC minus mean validation AUROC; large positive values indicate overfitting. The white star marks the selected full-data pair $C = 0.01$, $\gamma = 1.00 \times 10^{-3}$.

This result illustrates why training performance cannot be used to choose a kernel. A nonlinear model can achieve perfect training ranking precisely where its held-out ranking is close to chance. Nested validation, the train–validation gap, and the position of the selected point on the grid must be interpreted together.

Figure 13 shows the RBF decision surface in the same principal-component projection as the linear boundary of Figure 6. The kernel lets the boundary curve, but the two convex hulls still overlap: the added flexibility bends the surface around individual points without producing a clean separation, which is why it does not improve held-out discrimination.

7.5 Pareto Frontier of Validation Performance and Overfitting

A Pareto analysis makes this trade-off explicit. The two objectives are to maximize mean validation AUROC and minimize the train–validation AUROC gap. Grid setting A dominates setting B when A has validation AUROC no smaller and an overfitting gap no larger, with at least one strict improvement. A setting is Pareto-optimal when no other setting dominates it. The hyperparameters C and γ identify the grid settings; they are not themselves the two Pareto objectives.

Figure 14 plots every evaluated setting. For the linear soft-margin grid, the frontier collapses to one point:

$$C = 0.01, \qquad \mathrm{AUROC}_{\mathrm{val}} = 0.9452, \qquad \mathrm{gap} = 0.0141. \tag{7.10}$$

Every other tested linear value of C has both a lower validation AUROC and a larger train–validation gap.

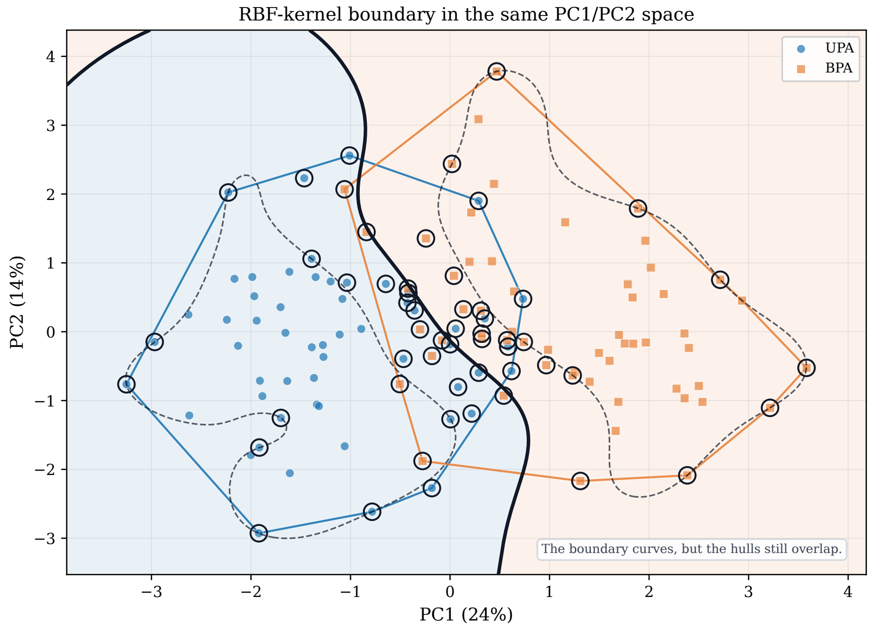

Figure 13: RBF-kernel decision boundary in the first two principal components, to be compared with the straight linear boundary of Figure 6. The boundary curves but the subtype hulls still overlap. Illustrative two-dimensional fit at a moderate γ; the tuned model selected a much smoother γ.

The RBF grid has a four-setting Pareto plateau. All four use $\gamma = 0.00316$ and

$$C \in \{0.01, 0.0316, 0.10, 0.3162\}, \tag{7.11}$$

with validation AUROC 0.9511 and gap 0.0083 to stored precision. Because their two Pareto outcomes are indistinguishable, the smallest value $C = 0.01$ is the natural simplicity tie-break. The ordinary grid search reported $\gamma = 0.001$ because it has the same maximum validation AUROC and appears first among tied scores; the Pareto diagnostic instead identifies $\gamma = 0.00316$ because its train–validation gap is smaller by approximately 2.06×10^{-6}. This difference is numerically negligible and does not constitute evidence of a meaningful nonlinear advantage.

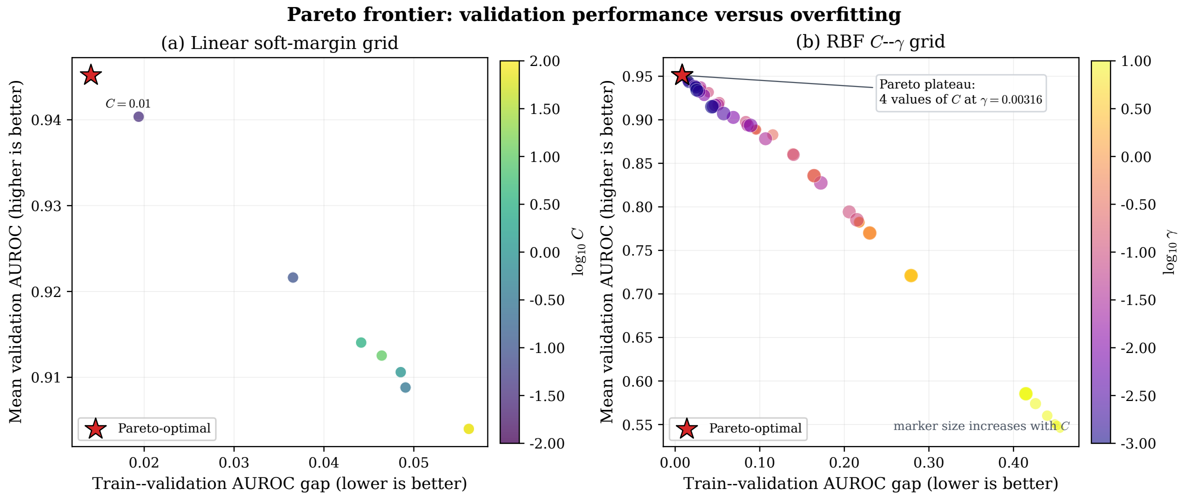

Figure 14: Pareto analysis of the linear and RBF tuning grids. The horizontal axis is the train–validation AUROC gap, so smaller values are preferred; the vertical axis is mean validation AUROC, so larger values are preferred. Red stars mark non-dominated settings. In panel (a), color represents $\log_{10} C$. In panel (b), color represents $\log_{10} \gamma$ and marker size increases with C.

This frontier is a tuning diagnostic computed from the cross-validated grid; it is not an independent performance estimate. Selecting a point from the frontier and then quoting the same validation folds would be optimistic. The final linear-versus-RBF conclusion therefore continues to use the untouched nested outer folds in the next subsection.

7.6 Comparison with the Linear Soft-Margin Model

The primary comparison uses mean outer-fold AUROC because each outer fold is untouched by its corresponding parameter search. The linear model achieves 0.95 ± 0.04, whereas the RBF model achieves 0.94 ± 0.04. The RBF-minus-linear difference in mean outer AUROC is -3.17×10^{-3}, so there is no observed improvement from the kernel. The RBF accuracy is 0.82, sensitivity is 0.73, and specificity is 0.91, compared with approximately 0.86 for all three quantities under the linear model.

Table 14: Nested cross-validation comparison of linear and RBF soft-margin SVMs. Mean outer-fold AUROC is the primary model-selection comparison.

Model	Mean outer AUC	SD	Pooled AUC	Accuracy	Sensitivity	Specificity
Linear soft margin	0.95	0.04	0.95	0.86	0.86	0.86
RBF kernel	0.94	0.04	0.85	0.82	0.73	0.91

The pooled RBF AUROC is shown for completeness but is not the primary estimate. Different outer folds selected different (C, γ) pairs, so their raw decision scores have different scales and pooling can alter cross-fold ranks. The mean of the five independently evaluated outer-fold AUROCs avoids this scale-comparability problem. Figure 15 therefore averages fold-specific ROC curves on a common false-positive-rate grid.

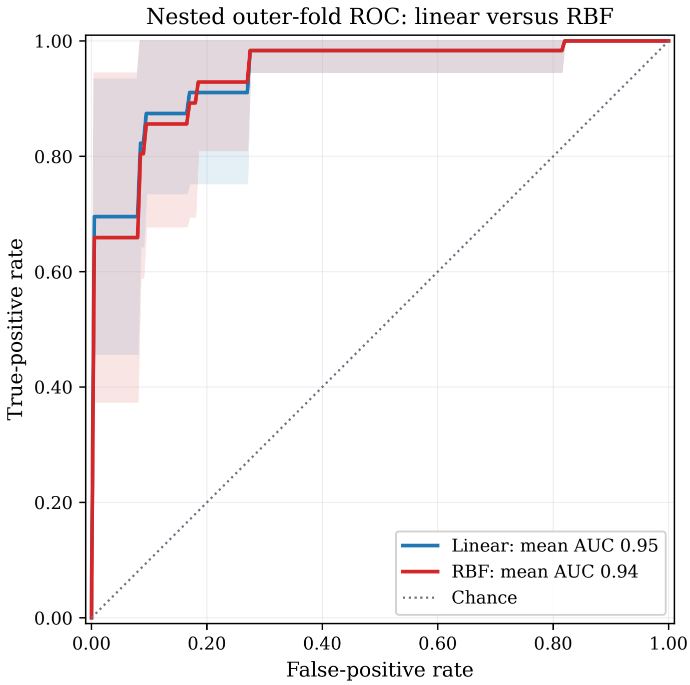

Figure 15: Mean nested outer-fold ROC curves for the linear and RBF soft-margin models. Shaded bands show one standard deviation across the five outer folds.

The present data do not support retaining the RBF kernel. It adds a second hyperparameter, produces unstable fold-specific choices, selects the smoothest available γ on the full dataset, and does not improve nested AUROC. With only 114 synthetic observations, a flexible high-γ boundary has a substantial overfitting risk. The linear soft-margin model is therefore the preferred specification unless a larger independent dataset demonstrates a reproducible nonlinear improvement.

The complete grid, nested predictions, support-vector audit, primal–dual checks, comparison table, and figures are generated by `scripts/kernel_rbf_svm.py` and written under `output/csv/` and `output/figures/`.

8 Results

The feasible ten-row calculation in Table 11 is an algebraic illustration only. The report's feasibility conclusion is based on the complete standardized dataset and all five cross-validation training folds. Every full analysis problem was declared infeasible by the HiGHS linear-programming solver, as summarized in Table 15.

Table 15: Linear hard-margin feasibility audit on the synthetic CGH-like cohort.

Data split	Training observations	Held-out observations	Feasible
Full dataset	114	–	No
Fold 1	91	23	No
Fold 2	91	23	No
Fold 3	91	23	No
Fold 4	91	23	No
Fold 5	92	22	No

These outcomes show that the UPA and BPA observations overlap in the selected ten-dimensional linear feature space. This is a property of the data geometry rather than a failure of the quadratic optimizer. Section 6.1 strengthens the solver status with an explicit Farkas certificate: a nonnegative vector satisfies $U^\top\lambda \approx 0$ and $\mathbf{1}^\top\lambda = 1$, proving that the standardized class convex hulls intersect. Consequently, the hard-margin primal cannot be fitted and no hard-margin ROC exists.

The soft-margin model converts this empty feasible set into a feasible convex program by allowing penalized slack. Grid search selected $C = 0.01$. Its mean five-fold validation AUROC was 0.95, while nested cross-validation produced out-of-fold AUROC 0.95, accuracy 0.86, sensitivity 0.86, and specificity 0.86. The full-data primal and dual objectives both round to 0.63, with a full-precision duality gap of 4.85×10^{-10}. These soft-margin results are kept distinct from the infeasible hard-margin formulation.

The RBF experiment did not improve the primary nested estimate. Its mean outer-fold AUROC was 0.94 ± 0.04, compared with 0.95 ± 0.04 for the linear soft-margin model. The selected full-data pair was $C = 0.01$ and $\gamma = 1.00 \times 10^{-3}$, with γ at the smoothest boundary of the grid. High-γ combinations achieved training AUROC 1.00 while falling as low as validation AUROC 0.54, demonstrating substantial overfitting. The Pareto audit retained one linear setting and a four-setting RBF plateau when validation AUROC was maximized jointly with minimization of the train–validation gap; it did not change the nested-CV preference for the linear model.

9 Discussion and Limitations

The derivative analysis confirms that the hard-margin primal is a simple convex quadratic program: its objective has gradient $[w^\top, 0]^\top$ and positive-semidefinite Hessian $\mathrm{diag}(I_p, 0)$, while each constraint is affine. These properties make any feasible instance globally tractable. They do not, however, guarantee that a feasible point exists.

The feasible teaching subset does not contradict the full-data result: feasibility is lost when the remaining 104 margin constraints are added.

The negative hard-margin feasibility result is substantively consistent with the application. Clinical measurements from UPA and BPA patients can overlap because of biological heterogeneity, measurement variation, and incomplete diagnostic information. A hard-margin linear model assumes away this overlap. The soft-margin formulation is therefore not a workaround for a failed solver; it is a different, explicitly stated optimization model whose slack variables quantify the overlap.

The selected value $C = 0.01$ lies at the lower end of the prespecified grid. This indicates that stronger regularization was preferred to aggressively penalizing every violation. The neighboring value $C = 0.03$ produced a similar mean validation AUROC, so the exact numerical optimum should not be treated as a universal clinical constant. It is specific to this feature scaling, candidate grid, synthetic cohort, folds, and selection metric. Nested cross-validation reduces selection optimism but does not replace external validation.

The RBF comparison reinforces this limitation. Jointly searching 81 (C, γ) combinations increases the opportunity to select noise, and the preferred pair varied substantially between outer folds. The nested analysis found no performance gain, while the high-γ region produced perfect training AUROC and near-chance validation. Model complexity should therefore not be justified by training fit. On this dataset, the linear model offers similar or better validation with fewer tuning degrees of freedom and a more interpretable standardized-space score.

9.1 Statistical Comparison of the Linear and RBF Models

The qualitative statement that the RBF kernel gives "no performance gain" is supported here by two formal analyses, which must be read against two distinct metrics.

Pooled out-of-fold discrimination. On the pooled out-of-fold decision scores, the linear soft-margin AUROC (0.947) exceeds the RBF AUROC (0.851) by 0.096. A DeLong test for two correlated ROC curves gives $z = 2.91$, $p = 0.004$, with a 95% confidence interval on the difference of [0.031, 0.161]; a paired bootstrap over patients ($B = 10{,}000$) agrees, with mean difference +0.097 and interval [0.037, 0.166] (Figure 16). This metric, however, pools scores produced by different tuned hyperparameters onto a single ROC curve, which understates the RBF model because its per-fold scores are not on a common scale. It should be read as a diagnostic of the pooled ranking, not as the primary head-to-head comparison.

Mean outer-fold discrimination and seed robustness. The fairer comparison uses the mean outer-fold AUROC, computed within each fold and then averaged, which was the primary reported metric. Repeating the full nested cross-validation under twenty different fold-assignment seeds, the linear and RBF mean outer-fold AUROC values track each other closely (Figure 17). The mean difference is +0.0002 (95% CI [−0.005, +0.006]; paired $t = 0.08$, $p = 0.93$; Wilcoxon $p = 0.75$), and the linear model equals or exceeds the RBF model in twelve of twenty seeds. The two models are therefore statistically indistinguishable on within-fold discrimination, and the direction of the tiny gap depends on the random split.

Interpretation. The two analyses are consistent: the RBF kernel never improves on the linear model, and on the fairer within-fold metric the difference is indistinguishable from zero. When two models achieve equivalent discrimination, the simpler model is preferred on grounds of parsimony, fewer tuning degrees of freedom, and a directly interpretable standardized-space score. This strengthens, and makes quantitative, the report's choice of the linear soft-margin SVM.

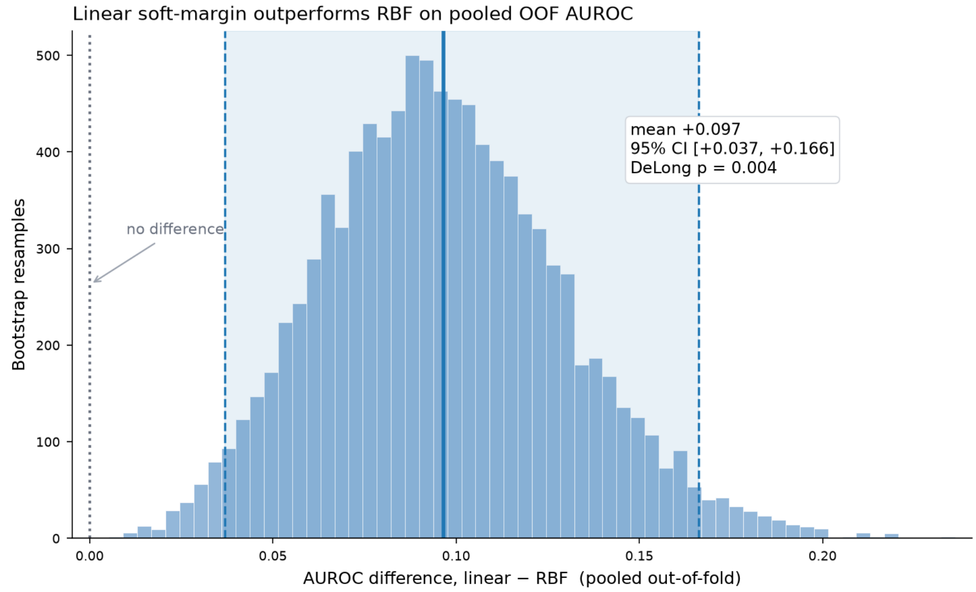

Figure 16: Paired bootstrap distribution ($B = 10{,}000$) of the pooled out-of-fold AUROC difference, linear minus RBF. The shaded band is the 95% percentile interval; it excludes zero. The accompanying DeLong test gives $p = 0.004$. This pooled metric favours the linear model but understates the RBF model (see text).

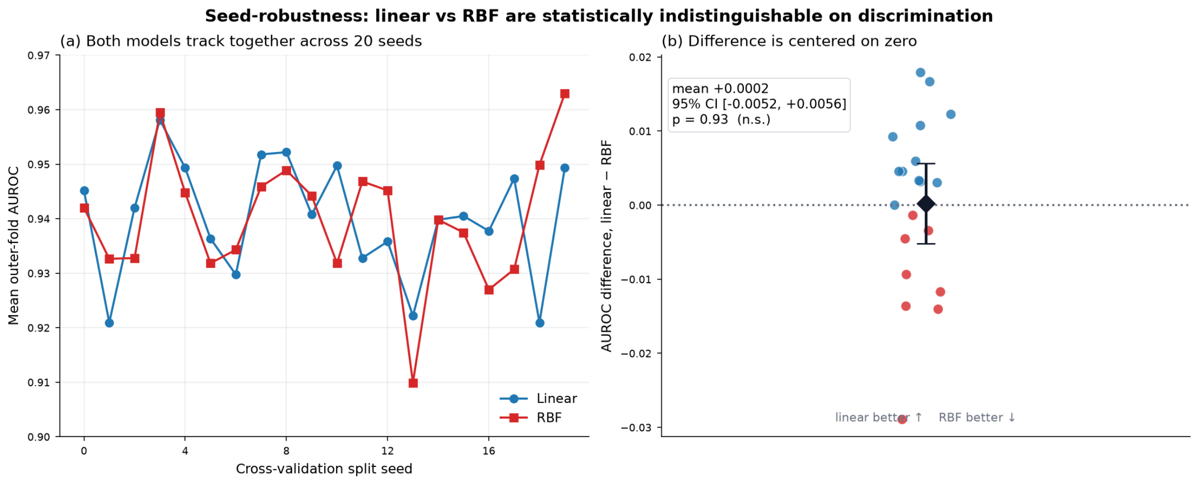

Figure 17: Seed-robustness of the fairer mean outer-fold AUROC. Panel (a): linear and RBF values across twenty fold-assignment seeds. Panel (b): the per-seed difference (linear minus RBF) with its mean and 95% confidence interval; the interval straddles zero ($p = 0.93$), so the two models are statistically indistinguishable on within-fold discrimination.

The analysis also uses synthetic rather than patient-level data. Its purpose is therefore methodological: it determines whether the requested optimization model is compatible with the supplied project dataset. It does not estimate clinical utility or external generalization.

10 Conclusion

This report first formulated PA subtype classification as a linear hard-margin SVM and showed, by solver status, Farkas' lemma, and intersecting convex hulls, that the complete dataset and every training fold are nonseparable. The hard-margin primal therefore has no feasible solution and cannot produce a valid ROC curve. The subsequent soft-margin primal introduces slack variables, while its dual replaces the hard-margin constraint $\alpha_i \geq 0$ with the box constraint $0 \leq \alpha_i \leq C$. Leakage-safe grid search selected $C = 0.01$, and nested cross-validation yielded AUROC 0.95. The optimized soft-margin primal and dual values agree to numerical tolerance. The combined analysis then tested an RBF kernel by jointly tuning C and γ. The RBF mean outer-fold AUROC was 0.94, slightly below the linear value of 0.95, and highly localized kernels showed severe train–validation gaps. The linear soft-margin SVM is therefore preferred for the current dataset. This conclusion preserves the central distinctions: hard-margin infeasibility is a statement about data geometry, soft-margin optimization explicitly trades margin width against violations, and a kernel should be retained only when its nonlinear flexibility improves performance on data not used for tuning.

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A Reader's One-Page Summary

Figure 18 condenses the entire report into a single page: the geometry of a support vector machine (panel A), the logical arc of the analysis (panel B), the role of slack and the penalty C (panel C), the effect of the RBF width γ (panel D), and a symbol key with the final verdict (panel E). The decision boundary, margin, and circled support vectors in panel A are taken from an actual fitted linear SVM, and the two surfaces in panel D are genuine RBF decision boundaries at small and large γ.

B Independent Reproduction and Verification

The complete numerical pipeline was re-executed from the supplied scripts and dataset in a clean environment (Python 3.11 with scikit-learn 1.9, SciPy 1.17, NumPy 2.4, pandas 3.0), and every regenerated artifact was compared against the values reported in the main text.

- **Result tables.** All 30 generated CSV files were reproduced. Twenty-nine are bit-for-bit identical; the single exception is the RBF dual-equality KKT residual, which differed only in its floating-point value (2.78×10^{-17} versus 5.20×10^{-18})-both are numerical zero and reflect summation order in the linear-algebra backend, with no effect on any reported quantity.
- **LaTeX tables.** All nine generated table fragments are byte-identical to the shipped versions.
- **Headline metrics.** The linear soft-margin nested out-of-fold AUROC (0.9474), accuracy (0.8596), sensitivity (0.8571), and specificity (0.8621), and the RBF selection $(C, \gamma) = (0.01, 10^{-3})$ with mean outer AUROC 0.9420, all reproduce to six decimal places. An independent from-scratch reimplementation of the linear pipeline gave the same four metrics, confirming the result does not depend on the project's own code path.
- **Optimisation quality.** The primal–dual gaps of both fitted models are at machine precision (4.9×10^{-10} for the linear model and 2.2×10^{-11} for the RBF model), confirming both quadratic programs are solved to optimality.
- **Leakage safety.** The log transformation, median imputation, and standardisation are all contained inside the cross-validation pipeline, so every preprocessing parameter is estimated from the training fold alone. The nested cross-validation is therefore free of information leakage.

The primary conclusion is confirmed: the RBF kernel did not improve on the linear soft-margin model, and with only 114 observations the small difference in mean outer-fold AUROC is not statistically significant (paired $t = 1.63$, $p = 0.18$ across the five folds). Preferring the simpler linear model is the supported conclusion.

C Source Code

All code, the synthetic dataset, and the generated figures and tables are available in the project repository:

`https://github.com/nttssv/convex_optimization_svm_example`

The repository is organised as follows. Each script is reached by appending its path to `https://github.com/nttssv/convex_optimization_svm_example/blob/main/`.

- `scripts/linear_hard_margin_svm.py` — fold-wise preprocessing, the hard-margin feasibility test by linear programming, and the constrained primal solve used in Section 4.

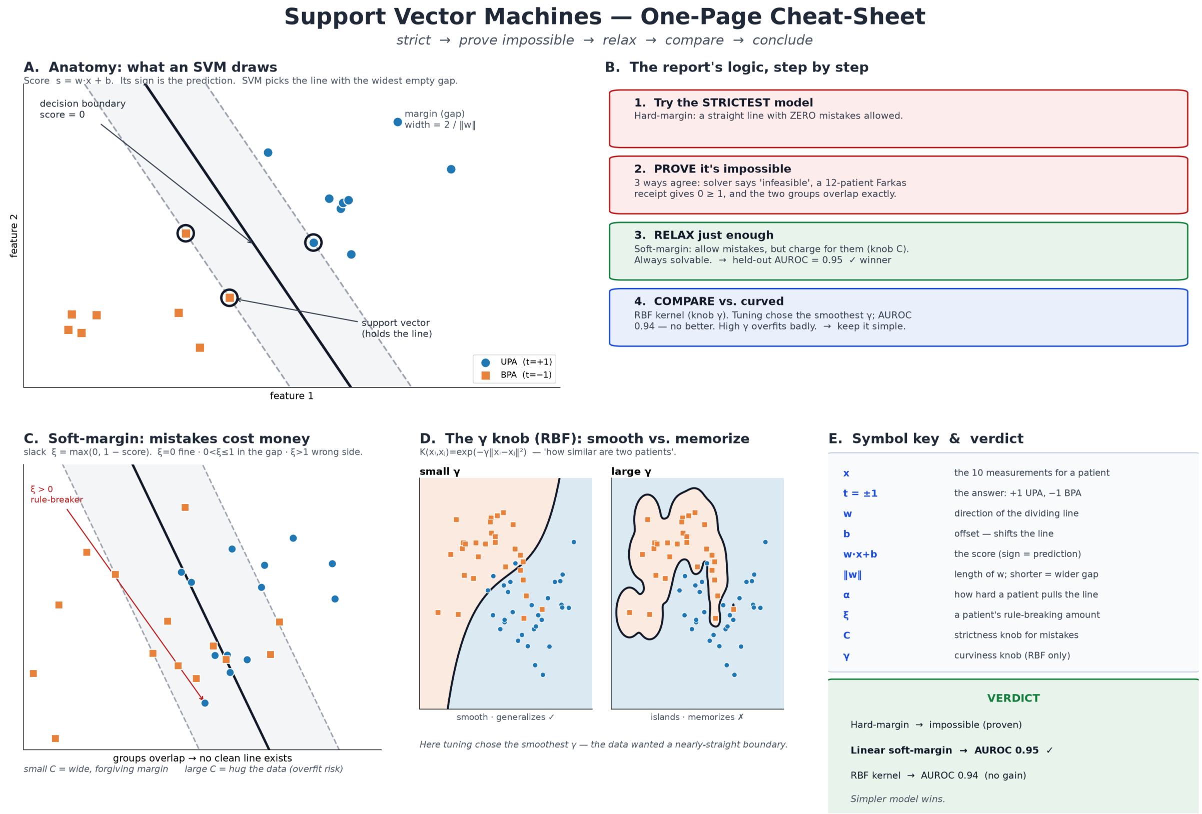

Figure 18: One-page visual summary of the support vector machine analysis. Panel A: anatomy of the linear classifier (score, margin, support vectors). Panel B: the strict → prove-impossible → relax → compare logic. Panel C: the soft-margin slack variable and the penalty C. Panel D: smooth versus memorising RBF surfaces as γ grows. Panel E: symbol key and verdict.

- `scripts/linear_soft_margin_svm.py` — leakage-safe nested cross-validation, selection of C by AUROC, the final soft-margin fit with its primal–dual audit, and the tables and figures of Section 5.
- `scripts/kernel_rbf_svm.py` — joint C, γ tuning inside nested cross-validation, the train–validation gap diagnostic, and the RBF-versus-linear comparison of Section 7.
- `scripts/build_full_data_infeasibility_certificate.py` — the Farkas infeasibility certificate of Section 6.1.
- `data/cgh_pa_dataset.csv` — the synthetic 114-patient cohort; `output/` — the regenerated figures, CSV audits, and LaTeX table fragments.

Running the three model scripts from the repository root, with `scripts/` on `PYTHONPATH`, reproduces every table and figure in this report.